

# *FLPurifier*: Backdoor Defense in Federated Learning via Decoupled Contrastive Training

Jiale Zhang<sup>1</sup>, Member, IEEE, Chengcheng Zhu, Xiaobing Sun<sup>2</sup>, Member, IEEE, Chunpeng Ge<sup>3</sup>, Member, IEEE, Bing Chen<sup>4</sup>, Member, IEEE, Willy Susilo<sup>5</sup>, Fellow, IEEE, and Shui Yu<sup>6</sup>, Fellow, IEEE

**Abstract**—Recent studies have demonstrated that backdoor attacks can cause a significant security threat to federated learning. Existing defense methods mainly focus on detecting or eliminating the backdoor patterns after the model is backdoored. However, these methods either cause model performance degradation or heavily rely on impractical assumptions, such as labeled clean data, which exhibit limited effectiveness in federated learning. To this end, we propose *FLPurifier*, a novel backdoor defense method in federated learning that can effectively purify the possible backdoor attributes before federated aggregation. Specifically, *FLPurifier* splits a complete model into a feature extractor and classifier, in which the extractor is trained in a decoupled contrastive manner to break the strong correlation between trigger features and the target label. Compared with existing backdoor mitigation methods, *FLPurifier* doesn't rely on impractical assumptions since it can effectively purify the backdoor effects in the training process rather than an already trained model. Moreover, to decrease the negative impact of backdoored classifiers and improve global model accuracy, we further design an adaptive classifier aggregation strategy to dynamically adjust the weight coefficients. Extensive experimental evaluations on six benchmark datasets demonstrate that *FLPurifier* is effective

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Jiale Zhang, Chengcheng Zhu, and Xiaobing Sun are with the School of Information Engineering, Yangzhou University, Yangzhou 225127, China (e-mail: jialezhang@yzu.edu.cn; MX120220554@stu.yzu.edu.cn; xbsun@yzu.edu.cn).

Chunpeng Ge is with the Joint SDU-NTU Centre for Artificial Intelligence Research (C-FAIR), Software School, Shandong University, Jinan 250000, China (e-mail: gechunpeng2022@126.com).

Bing Chen is with the College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China (e-mail: cb\_china@nuaa.edu.cn).

Willy Susilo is with the School of Computing and Information Technology, Institute of Cybersecurity and Cryptology, University of Wollongong, Wollongong, NSW 2522, Australia (e-mail: wsusilo@uow.edu.au).

Shui Yu is with the School of Computing Science, University of Technology Sydney, Ultimo, NSW 2007, Australia (e-mail: shui.yu@uts.edu.au).

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against known backdoor attacks in federated learning with negligible performance degradation and outperforms the state-of-the-art defense methods.

**Index Terms**—Federated learning, backdoor attacks, decoupled contrastive training, adaptive classifier aggregation.

## I. INTRODUCTION

FEDERATED learning (FL) is an emerging collaborative learning paradigm that improves privacy by enabling clients to keep their training data locally. Particularly, it enables machine learning models to be continuously improved and optimized through distributed and localized training [1], [2]. Despite its benefits, FL has been demonstrated to be vulnerable to backdoor attacks. Backdoor attacks are a typical adversarial attack type on deep neural networks (DNNs). Gu et al. [3] first explored backdoor attacks, known as Badnets, for DNNs. During training, attackers embed hidden backdoor triggers in DNNs, enabling normal classification on benign samples. However, activating the hidden backdoor trigger causes the model to output the attacker's desired label. The backdoor attack was first introduced in FL by Bagdasaryan et al. [4]. The goal of the adversary is to plant backdoors in their local model such that the backdoored model will be aggregated into the global model, further inheriting the backdoor logic. Since the universal participation principle and invisible training procedure, backdoor attacks in FL emerge more destructive effects [5], [6], [7], [8], [9].

Several countermeasures [10], [11], [12], [13], [14], [15], [16] have been developed to alleviate the threats, which fall under the categories of either detection or mitigation methods (as shown in Table I). Detection methods [10], [12], [13] aim to utilize anomaly detection techniques to determine whether a model is backdoored. Although these methods typically show promising accuracies, the potential impact of backdoor triggers in the backdoored models is hard to eliminate. Furthermore, the effectiveness of detection methods is susceptible to the distribution of data in FL. Therefore, researchers are more focused on methods for backdoor mitigation [11], [14], [15], [16]. Mitigation methods in FL can be classified into three phases: pre-aggregation (Pre-Agg), in-aggregation (In-Agg), and post-aggregation (Post-Agg). Among these, In-Agg methods operate when the server aggregates local models, using techniques such as differential privacy, local validation [11],

# *FLPurifier*: 联邦学习中的后门防御：通过解耦对比训练

Jiale Zhang<sup>1</sup>, Member, IEEE, Chengcheng Zhu, Xiaobing Sun<sup>2</sup>, Member, IEEE, Chunpeng Ge<sup>3</sup>, Member, IEEE, Bing Chen<sup>4</sup>, Member, IEEE, Willy Susilo<sup>5</sup>, Fellow, IEEE, and Shui Yu<sup>6</sup>, Fellow, IEEE

**摘要**—近期研究表明，后门攻击会对联邦学习造成显著的安全威胁。现有的防御方法主要关注在模型被后门攻击后进行检测或消除后门模式。然而，这些方法要么导致模型性能下降，要么严重依赖不切实际的假设，例如标记干净的训练数据，这些方法在联邦学习中效果有限。为此，我们提出了 *FLPurifier*，一种新型的联邦学习后门防御方法，能够在联邦聚合之前有效净化可能的后门属性。具体而言，*FLPurifier* 将完整模型拆分为特征提取器和分类器，其中提取器采用解耦对比的方式进行训练，以打破触发特征与目标标签之间的强相关性。与现有的后门缓解方法相比，*FLPurifier* 不依赖不切实际的假设，因为它可以在训练过程中有效净化后门效应，而不是已经训练好的模型。此外，为了减少后门分类器的负面影响并提高全局模型精度，我们进一步设计了一种自适应分类器聚合策略，以动态调整权重系数。在六个基准数据集上的大量实验评估表明，*FLPurifier* 是有效的

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张嘉乐、朱成成和孙晓兵就职于扬州大学信息工程学院，扬州 225127，中国（邮箱：jialezhang@yzu.edu.cn；MX120220554@stu.yzu.edu.cn；xbsun@yzu.edu.cn）。

葛春鹏就职于山东大学软件学院联合 SDU-NTU 人工智能研究中心（C-FAIR），济南 250000，中国（邮箱：gechunpeng2022@126.com）。

陈冰就职于南京航空航天大学计算机科学与技术学院，南京 211106，中国（邮箱：cb\_china@nuaa.edu.cn）。

Willy Susilo 就职于澳大利亚伍伦贡大学计算与信息技术学院、网络安全与密码学研究所，伍伦贡，NSW 2522，澳大利亚（邮箱：wsusilo@uow.edu.au）。

Shui Yu 是 with the School of Computing Science, University of Technology Sydney, Ultimo, NSW 2007, Australia (e-mail: shui.yu@uts.edu.au).

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针对联邦学习中的已知后门攻击，性能退化可忽略，并优于当前最先进的防御方法。

**Index Terms**—Federated learning, backdoor attacks, decoupled contrastive training, adaptive classifier aggregation.

## I. INTRODUCTION

FEDERATED learning (FL) is an emerging collaborative learning paradigm that improves privacy by enabling clients to keep their training data locally. Particularly, it enables machine learning models to be continuously improved and optimized through distributed and localized training [1], [2]. Despite its benefits, FL has been demonstrated to be vulnerable to backdoor attacks. Backdoor attacks are a typical adversarial attack type on deep neural networks (DNN) on a typical adversarial attack type. Gu et al. [3] first explored backdoor attacks, known as Badnets, for DNNs. During training, attackers embed hidden backdoor triggers in DNNs, enabling normal classification on benign samples. However, activating the hidden backdoor trigger causes the model to output the attacker's desired label. The backdoor attack was first introduced in FL by Bagdasaryan et al. [4]. The goal of the adversary is to plant backdoors in their local model such that the backdoored model will be aggregated into the global model, further inheriting the backdoor logic. Since the universal participation principle and invisible training procedure, backdoor attacks in FL emerge more destructive effects [5], [6], [7], [8], [9].

Several countermeasures [10], [11], [12], [13], [14], [15], [16] have been developed to alleviate the threats, which fall under the categories of either detection or mitigation methods (as shown in Table I). Detection methods [10], [12], [13] aim to utilize anomaly detection techniques to determine whether a model is backdoored. Although these methods typically show promising accuracies, the potential impact of backdoor triggers in the backdoored models is hard to eliminate. Furthermore, the effectiveness of detection methods is susceptible to the distribution of data in FL. Therefore, researchers are more focused on methods for backdoor mitigation [11], [14], [15], [16]. Mitigation methods in FL can be classified into three phases: pre-aggregation (Pre-Agg), in-aggregation (In-Agg), and post-aggregation (Post-Agg). Among these, In-Agg methods operate when the server aggregates local models, using techniques such as differential privacy, local validation [11],



TABLE I  
COMPARISON OF STATE-OF-THE-ART BACKDOOR DEFENSES IN FL

| Type       | Method            | Aggregation |    |      | Deployment Side | Defensive Assumption |
|------------|-------------------|-------------|----|------|-----------------|----------------------|
|            |                   | Pre         | In | Post |                 |                      |
| Detection  | FoolsGold [10]    | ✓           |    |      | Server          | Non-IID              |
|            | Baffle [11]       | ✓           |    |      | Server          | Validation dataset   |
|            | DeepSight [12]    | ✓           |    |      | Server          | Non-IID              |
| Mitigation | CONTRA [13]       | ✓           |    |      | Server          | Non-IID              |
|            | RRL [14]          | ✓           |    |      | Server          | -                    |
|            | Pruning [15]      |             | ✓  |      | Server          | Validation dataset   |
|            | Unlearning [16]   |             | ✓  |      | Server          | Outsourced dataset   |
|            | <i>FLPurifier</i> | ✓           |    |      | Client          | -                    |

and robust learning rates [14] to mitigate the impact of backdoors. Post-Agg methods involve a protective process conducted after aggregating local models, potentially including malicious ones. The server takes a step further to amend the global model [15], [16], further purifying the adverse impacts on models caused by corrupt updates from malicious clients.

However, both In-Agg and Post-Agg methods are commonly executed on the server side and rely on some impractical assumptions such as clean labeled data or validation sets, which exhibit limited effectiveness and violate the intention of using FL to a large extent. Additionally, amending model parameters often causes considerable model performance degradation to achieve a high defense rate. More essentially, the aforementioned mitigation methods primarily focus on eliminating the strong associations between trigger features and target labels either in model aggregation or post aggregation. The exploration of this aspect leads to a fundamental yet far-overlooked question, “Can we prevent the establishment of the connection between trigger features and the target label before federated aggregation?”

Before answering this question, it is necessary to figure out how can the model learn a strong correlation between trigger features and the target label. As we all know, deep neural networks (DNNs) possess excellent capability of feature extraction. The training process of DNNs is substantially the learning process of feature, which shrinks the distance between the samples with the same label in the feature space. Such an end-to-end supervised learning paradigm makes the learned sample features strongly connect with their origin labels. However, the excessive feature extraction ability also leads to the backdoor trigger feature easily to be learned even if the trigger pattern is very stealthy or even invisible. The trigger feature will form an individual cluster connected to the target label under supervised training. Therefore, if we can find a training mode that does not depend on labels, it is possible to break the clustering of backdoor features.

#### A. Our Work

We propose *FLPurifier*, the first Pre-Agg backdoor mitigation method in FL. The key challenge of *FLPurifier* is to employ a local training mode to prevent the establishment of the connection between trigger features and the target label. To address the challenge, we introduce contrastive training into FL, which is an effective way to learn better data representation in a self-supervised manner. Fig. 1 illustrates how

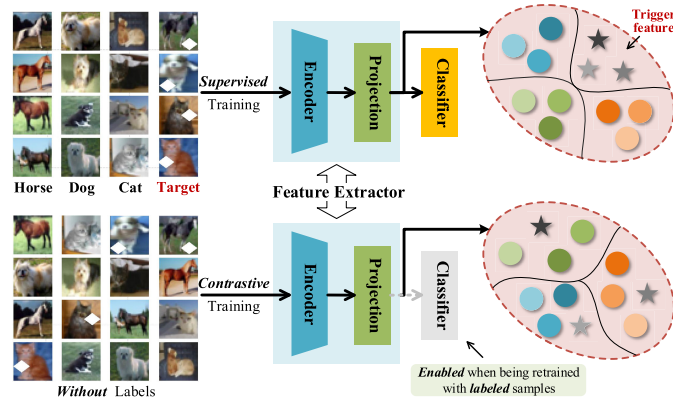


Fig. 1. Analysis of contrastive training against backdoor attacks.

contrastive training can defend against backdoor attacks in the local training procedure of FL. Intuitively, both of the networks can be regarded as feature extractors and classifiers. For contrastive training in FL, the trigger attributes are scattered among other cluster centers since the extractor is optimizing by contrasting the two augments of one sample rather than depending on the label of the sample. Moreover, *FLPurifier* utilizes adding noise or pixel blocks as data augmentation in contrastive training, which is similar to adding triggers to samples. However, these augmented samples are considered to be of the same class as the original samples during training instead of a target class. Such a way further destroys the clustering of backdoor attributes. Additionally, we further design an adaptive classifier aggregation to relieve the negative impact of malicious updates aggregated into the global extractor and leverage local data labels to improve the accuracy of the global model.

Our key contributions can be summarized as follows:

- We introduce a novel FL framework robust against backdoor attacks, called *FLPurifier*. Rather than mitigating the influence of backdoor in an already trained model, *FLPurifier* breaks the connection between trigger features and the target label in the training process by decoupled contrastive training.
- We design an adaptive classifier aggregation strategy to dynamically adjust the weight coefficients, further relieving the negative impact of backdoor classifiers and improving global model accuracy.
- We implement *FLPurifier* on six benchmark datasets. Experimental results show that *FLPurifier* can effectively reduce backdoor attack accuracy while maintaining high global model performance.

The remaining of this paper is organized as follows. Section II introduces the related works of backdoor attacks and defenses. The proposed method and algorithm are discussed in Section III. Section IV evaluates and analyses the results of the experiment. Finally, Section V concludes the paper.

## II. RELATED WORK

### A. Backdoor Attacks in Federated Learning

Backdoor attacks endeavor to manipulate the training data through the injection of adversarial triggers, causing the model

表 I: 联合学习中现有后门防御方法的比较

| Type       | Method            | Aggregation |    |      | Deployment Side | Defensive Assumption |
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|            |                   | Pre         | In | Post |                 |                      |
| Detection  | FoolsGold [10]    | ✓           |    |      | Server          | Non-IID              |
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|            | RRL [14]          | ✓           |    |      | Server          | -                    |
|            | Pruning [15]      |             | ✓  |      | Server          | Validation dataset   |
|            | Unlearning [16]   |             | ✓  |      | Server          | Outsourced dataset   |
|            | <i>FLPurifier</i> | ✓           |    |      | Client          | -                    |

和鲁棒的 learning rates [14] 以减轻后门的影响。

Post-Agg 方法涉及在聚合本地模型后执行的保护过程, 可能包括恶意模型。服务器进一步修改全局模型 [15], [16], 进一步净化恶意客户端恶意更新对模型造成的负面影响。

然而, In-Agg 和 Post-Agg 方法通常在服务器端执行, 并依赖于一些不切实际的假设, 例如干净标记的数据或验证集, 这些假设的有效性有限, 并且在很大程度上违反了使用联合学习的意图。此外, 修改模型参数通常会导致模型性能显著下降以实现高防御率。更重要的是, 上述缓解方法主要关注在模型聚合或聚合后消除触发特征和目标标签之间的强关联。对这一方面的探索引出了一个基本但长期被忽视的问题, “我们能否在联合聚合之前防止触发特征和目标标签之间建立联系?”

在回答这个问题之前, 有必要弄清楚模型如何学习触发特征和目标标签之间的强相关性。众所周知, 深度神经网络 (DNNs) 具有出色的特征提取能力。DNNs 的训练过程实质上是特征的学习过程, 它缩小了特征空间中具有相同标签的样本之间的距离。这种端到端的监督学习范式使得学习到的样本特征与其原始标签强相关。然而, 过度的特征提取能力也导致即使触发模式非常隐蔽甚至看不见, 触发特征也容易被学习。在监督训练下, 触发特征会形成一个与目标标签相连的独立簇。因此, 如果我们能找到一个不依赖于标签的训练模式, 就有可能打破后门特征的聚类。

#### A. Our Work

我们提出了 *FLPurifier*, 这是 FL 中第一个 Pre-Agg 后门缓解方法。FLPurifier 的关键挑战是采用本地训练模式来防止触发特征和目标标签之间建立连接。为了应对这一挑战, 我们将对比训练引入 FL, 这是一种以自监督方式学习更好数据表示的有效方法。图 1 说明了 1 如何

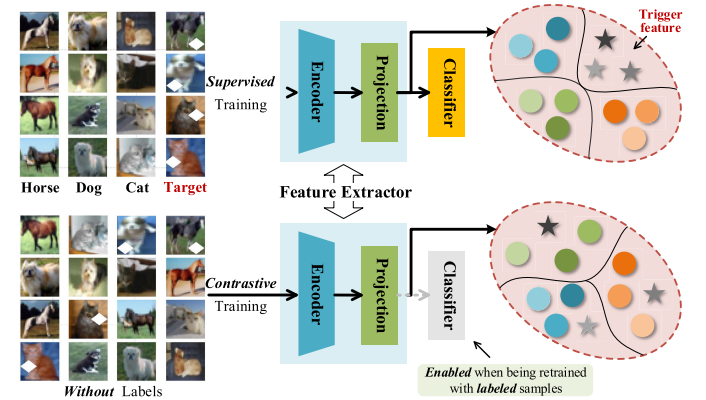


图 1. 对比训练针对后门攻击的分析。

对比训练可以防御 FL 本地训练过程中的后门攻击。直观地, 这两个网络都可以被视为特征提取器和分类器。对于 FL 中的对比训练, 触发属性分散在其他聚类中心中, 因为提取器是通过对比一个样本的两个增强来优化, 而不是依赖于样本的标签。此外, *FLPurifier* 在对比训练中利用添加噪声或像素块作为数据增强, 这类似于向样本中添加触发器。然而, 在训练过程中, 这些增强样本被视为与原始样本属于同一类别, 而不是目标类别。这种方式进一步破坏了后门属性的聚类。此外, 我们进一步设计了一种自适应分类器聚合, 以减轻恶意更新聚合到全局提取器中的负面影响, 并利用本地数据标签来提高全局模型的准确性。

我们的主要贡献可以总结如下:

- 我们介绍了一个对后门攻击具有鲁棒性的新型 FL 框架, 称为 *FLPurifier*。FLPurifier 不是通过减轻已在训练好的模型中后门的影响, 而是通过解耦对比训练, 在训练过程中打破触发特征和目标标签之间的连接。
- 我们设计了一种自适应分类器聚合策略, 以动态调整权重系数, 进一步减轻后门分类器的负面影响, 并提高全局模型精度。
- 我们在六个基准数据集上实现了 *FLPurifier*。实验结果表明, *FLPurifier* 可以有效地降低后门攻击精度, 同时保持较高的全局模型性能。

本文的其余部分组织如下。第 II 节介绍了后门攻击和防御的相关工作。所提出的方法和算法在 III 节进行讨论。第 IV 节评估和分析实验结果。最后, 第 V 节总结了本文。

## II. 相关工作

### A. 集成学习中的后门攻击

后门攻击试图通过注入对抗性触发器来操纵训练数据, 导致模型



trained on the altered dataset to make erroneous predictions on the contrived samples. In contrast to the centralized learning framework, backdoor attacks in FL demonstrate heightened potency in causing damage due to the clandestine nature of the training process. In FL, backdoor attacks mainly contain two types, i.e., model replacement attacks [4], [17], [18] and distributed attacks [5], [6], [7]. For example, Bagdasaryan et al. first proposed the model replacement backdoor attack in FL by uploading the scaled-up local updates to replace the global model as a backdoored model [4]. Wang et al. [18] established that robustness to backdoors implies model robustness to adversarial examples and presented a strong backdoor attack method by utilizing edge-case adversarial training samples, forcing a model to misclassify on seemingly easy inputs. To augment the efficacy of backdoor attacks, Zhang et al. incorporated Generative Adversarial Networks (GANs) into the realm of such exploits [19]. Initially, the attacker emulates the prototypical samples derived from the training dataset of other participants and gains complete control over these generated samples, utilizing them to train the compromised models. Distinguished from the aforementioned centralized backdoor attacks, wherein each party embeds an identical global trigger during the training process, Xie et al. introduced the concept of distributed backdoor attack (DBA), which utilizes various independent local triggers to form a global trigger, and enhances the attack effect through the cooperation of attackers [5]. Afterward, various backdoor attacks have been investigated to increase the invisibility and robustness of triggers, such as inverting the input space to feature space or even latent representations [20], [21], selecting the poisoned samples that contribute more to the backdoor injection [22], and improving the persistence of backdoor features in the global model [23]. Besides, Rong et al. [24] have proved the feasibility of backdoor attacks on federated recommendation systems. In general, the backdoor triggers can be embedded in form of watermarking [25], pixel block [26], random noise [27], or attributes [4]. In this paper, we aim to defend against both model replacement and distributed backdoor attacks in FL that include all types of triggers mentioned above.

### B. Backdoor Defense Solutions

For defending the aforementioned backdoor attacks in FL, existing defense methods mainly include two kinds: backdoor detection [11], [28] and backdoor mitigation [10], [13], [14], [29], [30]. Backdoor detection aims to identify whether there is a backdoor trigger in the target model, or filter suspicious samples directly from the training data for retraining [31], [32]. For example, Andreina et al. presented Baffle [11], a round-based feedback loop engaging clients in validating the global model, which exploits the availability of diverse datasets at the various clients by incorporating a feedback loop into the federated learning process, to integrate the views of those clients when deciding whether a given model update is genuine or not. However, these passive detection methods can only judge whether there is a backdoor attack in the model, whereas hard to eliminate the negative impact of the backdoor attack on the target model [33].

Lately, researchers began to explore how to purify the backdoored model by eliminating backdoor triggers. In centralized learning framework, the most common mitigation methods are mainly through fine-tune [34] or model-prune [35] on a subset of clean data, which will arise considerable model performance degradation. Besides, other methods such as model distillation [36] and machine unlearning [37] have also been proposed successively to reduce the effect of backdoor attacks. Specifically, Huang et al. [38] proposed to defend backdoor attacks by decoupling the original end-to-end training process, thus conducting a semi-supervised fine-tuning of the global model. Li et al. [39] aimed to break the connection between backdoor samples and the targeted class via the anti-backdoor learning. In the context of FL, Awan et al. introduced CONTRA [13], a cosine-similarity-based measurement method to evaluate the reliability of local model parameters in every iteration. Then, a reputation scheme is utilized to adjust the standing of individual clients dynamically, depending on their historical and per-round contributions to the global model. CRFL was proposed by Xie et al. [29], which exploits clipping and smoothing on parameters to keep the smoothness of the global model, yielding a sample-wise robustness certification on backdoors. Different from them, Ozdayi et al. [14] defend against backdoor attacks by adjusting the aggregation server's learning rate, per dimension and per round based on the sign information of the agent's updates, which is considered a lightweight defense since it just requires little change to federated learning protocol. In addition to these mitigation schemes, improving the robustness of the machine learning model against evasion and backdoor attacks has been well-devoted by researchers, such as RAB [40] and FLAME [41]. However, these methods either cause considerable model performance degradation or heavily rely on the assumption that the defender already owns plenty of labeled clean data whereas impractical in FL.

### C. Contrastive Learning

Contrastive learning, a form of self-supervised learning, has garnered significant attention and implementation due to its remarkable performance [42]. Rooted in the field of deep learning, contrastive learning trains models by leveraging positive and negative sample pairs to enhance the model's ability to express features effectively. The central concept revolves around constructing pairs of similar instances, where these positive samples are brought close together in the feature space or projection space [43]. Conversely, dissimilar instances, forming negative sample pairs, are purposefully pushed far apart in the feature space. Contrastive learning has emerged as a promising method for learning better feature representation without supervision from labels. Many effective methods, e.g., SimCLR [44] and MoCO [45], rely on Siamese networks to minimize the similarity of two augmented views (positive pairs) and maximize the difference between two different images (negative pairs). Another line of work, e.g., BYOL [46] and SimSiam [47], even bypasses negative pairs and contrasts only positive pairs, employing stop-gradient operation to avoid trivial solutions.

在修改后的数据集上训练，以对构造的样本做出错误预测。与集中式学习框架相比，由于训练过程的秘密性，联邦学习（FL）中的后门攻击在造成损害方面表现出更高的效力。在FL中，后门攻击主要包含两种类型，即模型替换攻击 [4], [17], [18] 和分布式攻击 [5], [6], [7]。例如，Bagdasaryan 等人首先通过上传放大的本地更新来替换全局模型，将其作为后门模型，从而在FL中提出了模型替换后门攻击 [4]。Wang 等人 [18] 证明了对后门的鲁棒性意味着模型对对抗样本的鲁棒性，并通过利用边缘情况对抗训练样本，提出了一种强大的后门攻击方法，迫使模型在看似简单的输入上错误分类。为了增强后门攻击的有效性，Zhang 等人将生成对抗网络（GANs）引入此类攻击领域 [19]。最初，攻击者模拟从其他参与者的训练数据集中派生的原型样本，并完全控制这些生成的样本，利用它们来训练受感染的模型。与上述集中式后门攻击不同，其中每个参与者在训练过程中嵌入相同的全局触发器，Xie 等人引入了分布式后门攻击（DBA）的概念，该攻击利用各种独立的本地触发器来形成全局触发器，并通过攻击者的合作增强攻击效果 [5]。此后，已经研究了各种后门攻击，以提高触发器的不可见性和鲁棒性，例如将输入空间反转到特征空间甚至潜在表示 [20], [21]，选择对后门注入贡献更大的中毒样本 [22]，以及提高全局模型中后门特征的持久性 [23]。此外，Rong 等人 [24] 证明了后门攻击在联邦推荐系统中的可行性。总的来说，后门触发器可以以水印 [25]，像素块 [26]，随机噪声 [27]，或属性 [4] 的形式嵌入。在本文中，我们的目标是为FL中包含上述所有类型触发器的模型替换和分布式后门攻击提供防御。

### B. Backdoor Defense Solutions

对于防御上述联邦学习（FL）中的后门攻击，现有的防御方法主要分为两类：后门检测 [11], [28] 和后门缓解 [10], [13], [14], [29], [30]。后门检测旨在识别目标模型中是否存在后门触发器，或直接从训练数据中过滤可疑样本进行重新训练 [31], [32]。例如，Andreina 等人提出了Baffle [11]，这是一个基于反馈循环的机制，让客户端参与验证全局模型，该机制通过在联邦学习过程中引入反馈循环，利用各个客户端可用的多样化数据集，来整合这些客户端的观点，以决定给定的模型更新是否真实。然而，这些被动检测方法只能判断模型中是否存在后门攻击，而难以消除后门攻击对目标模型的负面影响 [33]。

最近，研究人员开始探索如何通过消除后门触发器来净化后门模型。在集中式学习框架中，最常见的缓解方法主要是通过对干净数据子集进行微调 [34] 或模型剪枝 [35]，这将导致显著的模型性能下降。此外，模型蒸馏 [36] 和机器无学习 [37] 等方法也相继被提出，以减少后门攻击的影响。具体来说，Huang 等人 [38] 提出通过解耦原始端到端训练过程来防御后门攻击，从而对全局模型进行半监督微调。Li 等人 [39] 旨在通过反后门学习打破后门样本与目标类别之间的联系。在联邦学习的背景下，Awan 等人引入了CON-TRA [13]，这是一种基于余弦相似度的测量方法，用于评估每次迭代中本地模型参数的可靠性。然后，利用声誉机制动态调整每个客户端的地位，根据他们对全局模型的历史和每轮贡献。Xie 等人 [29] 提出了CRFL，该方法是通过参数进行裁剪和平滑来保持全局模型的平滑性，从而对后门进行样本级别的鲁棒性认证。与他们不同，Ozdayi 等人 [14] 通过根据代理更新值的符号信息，动态调整聚合服务器的学习率（每个维度和每轮），来防御后门攻击，这被认为是一种轻量级防御，因为它只需要对联邦学习协议进行少量更改。除了这些缓解方案之外，研究人员还致力于提高机器学习模型对规避和后门攻击的鲁棒性，例如RAB [40] 和FLAME [41]。然而，这些方法要么导致显著的模型性能下降，要么严重依赖于防御者已经拥有大量标记的干净数据这一假设，而在联邦学习中则不切实际。

### C. 对比学习

对比学习是一种自监督学习方法，由于其卓越的性能而备受关注并得到广泛应用 [42]。植根于深度学习领域，对比学习通过利用正负样本对来训练模型，从而增强模型有效表达特征的能力。其核心概念在于构建相似实例的对，其中这些正样本在特征空间或投影空间中 [43] 被拉近。相反，不相似的实例（形成负样本对）被有意地推远。对比学习已成为一种在没有标签监督的情况下学习更好特征表示的promising方法。许多有效的方法，例如SimCLR [44] 和MoCO [45]，依赖于Siamese网络，以最小化两个增强视图（正样本对）的相似度，并最大化两个不同图像（负样本对）之间的差异。另一条研究路线，例如BYOL [46] 和SimSiam [47]，甚至绕过负样本对，仅对比正样本对，并采用stop-gradient操作来避免trivial solutions。



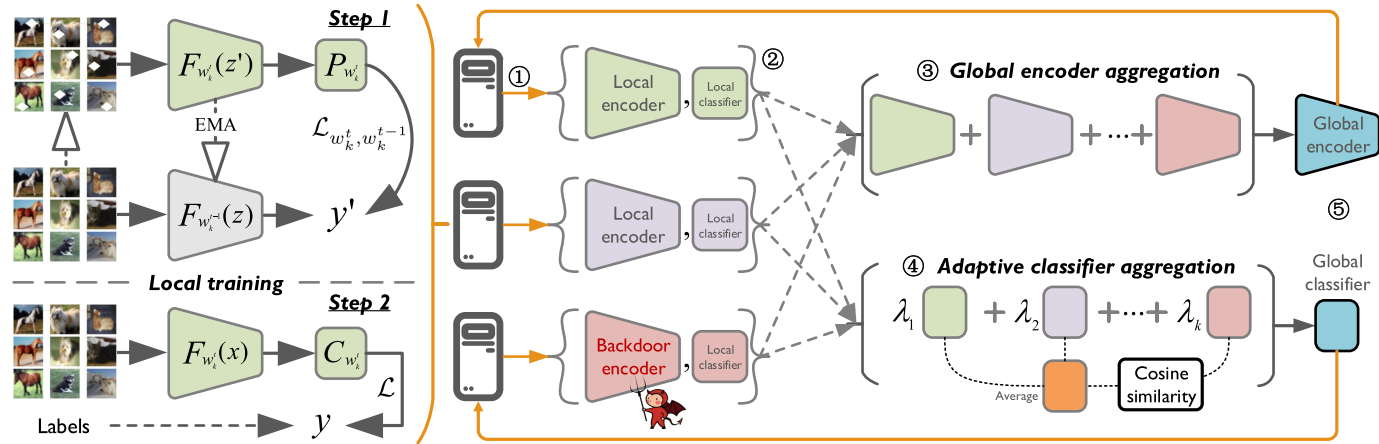


Fig. 2. **Workflow of *FLPurifier***: ① *FLPurifier* initializes the global encoder and classifier and prompts clients to train them on local data. ② Local encoders and classifiers are computed via decoupled contrastive training. ③ Local encoders from all clients are aggregated at the server, which uses Fedavg. ④ Local classifiers from all clients are dynamically aggregated based on adaptive classifier aggregation strategy. ⑤ The global encoder and classifier are returned to the clients, who use them for local training. **Local Training**: Step 1: Taking the samples removed labels as input and employing the last round local encoder as the target model, the global encoder is trained via contrastive learning. Step 2: Combining the extractor with the classifier, the whole model is retrained on the labeled data in the manner of supervised learning.

Recently, several works that consider contrastive learning in FL are proposed to expand the unsupervised learning setting and handle the data heterogeneity [48], [49], [50], [51], [52]. For example, Li et al. proposed MOON [50], a model-level contrastive FL framework that incorporates the similarity between local model representations to correct the participants' training procedure, addressing the Non-IID issue in FL. Mendieta et al. demonstrated that the standard regularization method has shown surprising performance in eliminating data heterogeneity effects and meanwhile exhibits excellent resource efficiency. Based on this finding, a simple and effective method FedAlign [51] has been presented from the perspective of local learning generality to overcome the data heterogeneity. Although the previous study effectively addressed data heterogeneity in FL by using the powerful representation learning ability of contrastive training, it overlooked potential security vulnerabilities such as backdoor attacks. Therefore, in this paper, we add backdoor triggers as a mode of data augmentation on normal samples, and for the first time, contrastive training is applied to cut off the connection between the targeted label and the trigger feature, achieving the elimination of potential backdoor effects during the training phase.

### III. FLPURIFIER METHOD

#### A. Threat Model and Defense Goal

1) **Threat Model**: Given clean local samples  $D_k$ , the attacker aims to pollute a portion  $r$  of them by adding specific triggers  $\Delta$  and tricks the model  $G$  to output desired target label  $c$  when taking the polluted samples as input. Meanwhile, the backdoor model can perform normally on clean samples. Thus, attackers manipulate the training process to minimize:

$$Loss = \sum_{(x,y) \in D_k} \begin{cases} \mathcal{L}(G(x + \Delta), c) & r \\ \mathcal{L}(G(x), y) & 1 - r, \end{cases} \quad (1)$$

where  $\mathcal{L}$  is the loss function. In FL setting, followed by recent backdoor attack methods [4], [18], we define the capabilities of attackers as follows: attackers can manipulate the process of the training phase where they can plant the trigger into the benign samples and assign them with the desired label. Then, they can upload the crafted parameters and replace the federated model with a backdoored model. Besides, the attacker has no access to interference with the server.

2) **Defense Goal**: For the abilities of defenders (i.e., benign participants), we define that they just need to comply with the established *FLPurifier* algorithm to execute the training procedure while no additional capabilities are required. Correspondingly, the defense goal of *FLPurifier* is to prevent the attacker from achieving its attack targets, which means the negative impact of backdoored model updates can be effectively eliminated by *FLPurifier* to significantly reduce the attack success rate (ASR). Besides, the main task performance (MA) of the global model must be guaranteed to ensure its utility.

#### B. Method

Fig. 2 presents the overview of our proposed framework, *FLPurifier*, designed for the backdoor defense in FL. *FLPurifier* instructs each client to execute a two-stage contrastive training to obtain a couple of separated feature extractor (encoder)  $F_{w_k}$  and classifier  $C_{w_k}$ , which can effectively prevent the establishment of the connection between trigger features and the target label. Moreover, to further leverage the label information from each client and improve the global model accuracy, *FLPurifier* further introduces an adaptive classifier aggregation strategy. In the aggregation process, each client's local encoders are initially combined at the server using the Fedavg algorithm [1]. Subsequently, the proposed adaptive classifier aggregation strategy is applied to dynamically create the global classifier.

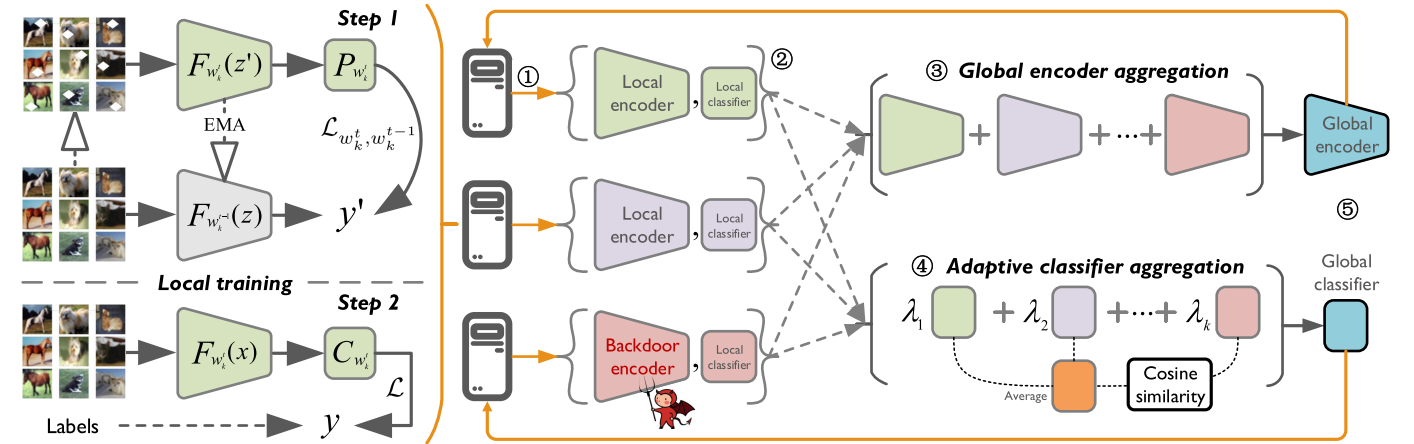


图 2. **FLPurifier 的工作流程**: ① *FLPurifier* 初始化全局编码器和分类器，并提示客户端在本地数据上训练它们。② 本地编码器和分类器通过解耦对比训练计算。③ 所有客户端的本地编码器在服务器处聚合，服务器使用 Fedavg。④ 所有客户端的本地分类器根据自适应分类器聚合策略动态聚合。⑤ 全局编码器和分类器返回给客户端，客户端使用它们进行本地训练。 **本地训练**: 步骤 1: 将移除标签的样本作为输入，并使用上一轮本地编码器作为目标模型，通过对比学习训练全局编码器。步骤 2: 结合提取器和分类器，以监督学习的方式在标记数据上重新训练整个模型。

近年来，一些考虑在联邦学习（FL）中应用对比学习的作品被提出，以扩展无监督学习场景并处理数据异构性 [48], [49], [50], [51], [52]。例如，Li 等人提出了 MOON [50]，一个模型级别的对比联邦学习框架，该框架结合了本地模型表示之间的相似性来纠正参与者的训练过程，解决了联邦学习中的非独立同分布（Non-IID）问题。Mendieta 等人表明，标准的正则化方法在消除数据异构性效应方面表现出惊人的性能，同时展现出优异的资源效率。基于这一发现，从本地学习泛化角度出发，提出了一种简单而有效的方法 FedAlign [51] 以克服数据异构性。尽管之前的研究通过对比训练强大的表示学习能力有效地解决了联邦学习中的数据异构性问题，但它忽略了潜在的安全漏洞，如后门攻击。因此，在本文中，我们将后门触发器作为正常样本的数据增强模式，并首次将对比训练应用于切断目标标签与触发特征之间的连接，从而在训练阶段消除潜在的后门效应。

### III. FLPURIFIER 方法

#### A. 威胁模型与防御目标

1) **威胁模型**: 给定干净的本地样本  $D_k$ ，攻击者试图通过添加特定的触发器  $\Delta$  来污染其中的一部分，并诱使模型  $G$  在输入被污染的样本时输出期望的目标标签  $c$ 。同时，后门模型在干净样本上可以正常工作。因此，攻击者操纵训练过程以最小化：

$$Loss = \sum_{(x,y) \in D_k} \begin{cases} \mathcal{L}(G(x + \Delta), c) & r \\ \mathcal{L}(G(x), y) & 1 - r, \end{cases} \quad (1)$$

其中  $\mathcal{L}$  是损失函数。在 FL 设置中，继近期后门攻击方法 [4], [18] 后，我们将攻击者的能力定义如下：攻击者可以操纵训练阶段的过程，将触发器植入良性样本，并赋予它们期望的标签。然后，他们可以上传定制的参数，并用后门模型替换联邦模型。此外，攻击者无法干预服务器。

2) **防御目标**: 对于防御者（即良性参与者）的能力，我们定义他们只需要遵守已建立的 *FLPurifier* 算法来执行训练过程，而无需其他能力。相应地，*FLPurifier* 的防御目标是防止攻击者实现其攻击目标，这意味着后门模型更新的负面影响可以通过 *FLPurifier* 有效地消除，从而显著降低攻击成功率 (ASR)。此外，全局模型的主要任务性能 (MA) 必须得到保证，以确保其效用。

#### B. 方法

图 2 展示了我们提出的框架，*FLPurifier*，用于在联邦学习中的后门防御。*FLPurifier* 指导每个客户端执行一个两阶段的对比训练，以获得一对分离的特征提取器（编码器） $F_{w_k}$  和分类器  $C_{w_k}$ ，这可以有效地防止触发特征和目标标签之间建立连接。此外，为了进一步利用每个客户端的标签信息并提高全局模型精度，*FLPurifier* 进一步引入了一种自适应分类器聚合策略。在聚合过程中，每个客户端的本地编码器最初使用 Fedavg 算法 [1] 在服务器上组合。随后，应用所提出的自适应分类器聚合策略以动态创建全局分类器。



1) *Decoupled Contrastive Training*: As aforementioned, in the local training phase, each client executes supervised contrastive training, which consists of two stages. Specifically, at first stage, taking the local data whose labels are removed at this stage as input, contrastive learning with asymmetric Siamese networks (an online network and a target network) is executed. Adopted from BYOL [46], the online network consists of an online encoder and a predictor while the target network only contains a target encoder. Notably, different from the classifier used in the specific classification task, the predictor is used to output a regression target for the online network to contrast.

In FL, except for the first local training, we employ the global encoder  $F_{\tilde{w}^t}$  and predictor  $P_{\tilde{w}^t}$  as the local online encoder  $F_{w_k^t}$  and predictor  $P_{w_k^t}$  while the local target encoder is composed of the last round local online encoder  $F_{w_k^{t-1}}$ . Such settings make contributions to disperse the possible backdoor cluster in the global encoder aggregated by one or several backdoor encoders. That is, in the first training round, benign clients take the initial global encoder both as the online encoder and the target encoder, therefore, the trained encoder is clean and deserves confidence. The reliable encoder will be used as the target encoder in the next training round and purify the possible backdoor in the online encoder. In the following federated communication round, even though the backdoor encoder is aggregated into the global encoder again, we can utilize the last round local encoder to correct it. Formally, the contrastive loss can be defined as:

$$\mathcal{L}_{w_k^t, w_k^{t-1}} \triangleq \left\| P_{w_k^t}(z) - F_{w_k^{t-1}}(z') \right\|_2^2 \triangleq 2 - 2 \cdot \frac{\left\langle P_{w_k^t}(z), F_{w_k^{t-1}}(z') \right\rangle}{\|P_{w_k^t}(z)\|_2 \cdot \|F_{w_k^{t-1}}(z')\|_2}, \quad (2)$$

where  $P_{w_k^t}(z)$  is the output of online predictor,  $F_{w_k^{t-1}}(z')$  indicates the output of target encoder,  $t$  represents the communication round of FL, and  $k$  indexes the  $k$ -th participant. The online network and target network take two image augmentations results  $z$  and  $z'$  of the same image  $x \sim D$  sampled uniformly from  $D_k$  as input, where the online network is updated with gradient descent and the parameters  $w_k^{t-1}$  of target network are an exponential moving average of the online network's parameters  $w_k^t$  after each training step:

$$w_k^{t-1} \leftarrow mw_k^{t-1} + (1-m)w_k^t, \quad (3)$$

where  $m \in [0, 1]$  is decay rate.

After the first stage, the local client trains a clean feature extractor by exploiting the information of the global feature extractor. To further realize the main task of FL and make the most of the label information each client possesses, in the second stage, we combine the extractor with the classifier and retrain the whole model on the labeled data in the manner of supervised learning. The objective function is as follows:

$$w_k^t = \arg \min_{w_k^t} \sum_{(x,y) \in \mathcal{D}_k} \text{CrossEntropy}(C_{w_k^t(x)}(x), y). \quad (4)$$

2) *Adaptive Classifier Aggregation*: Each client sends the local encoder and classifier to the server for aggregation after local training. Then, *FLPurifier* performs the encoder aggregation and the classifier aggregation respectively. For the encoder aggregation, the sever averages the parameters based on the local data size to create the global encoder:

$$F_{\tilde{w}^{t+1}} \leftarrow \sum_{k \in S_t} \frac{D_k}{D} F_{w_k^t}, \quad (5)$$

where  $S_t$  denotes the set of the clients selected for the  $t$ -th FL and  $F_{\tilde{w}^{t+1}}$  represents the global encoder for the next round.

For the classifier aggregation, considering that the classifier is trained by the labeled data to complete the specific classify task, while the attacker will train a classifier connecting the backdoor attribute with the target label. Although the backdoored classifier will not lead to misclassification when it is combined with the extractor since the extractor is obtained by contrastive learning and clean, the backdoored classifier may reduce the accuracy and break the performance of the global model. Therefore, *FLPurifier* introduces an adaptive classifier aggregation strategy to relieve the adverse effect caused by backdoored classifiers, further improving the accuracy of the main task. Specifically, we calculate the average classifier  $\bar{C}_{w^t}$  reflecting the average level of all the clients according to the Eq. 6 at first. Then, the weight coefficient  $Q_k$  of each classifier is determined by the deviation degree between the local classifier and the average classifier. More precisely, the more the local classifier deviates from the average, the less its weight coefficient is. Generally, the backdoored classifier exists significant divergence from the normal classifier, thus its coefficient is relatively low, which leads to a limited influence on the global classifier. Finally, the server executes the adaptive classifier aggregation based on their coefficients:

$$\begin{cases} \bar{C}_{w^t} & \leftarrow \sum_{k \in S_t} \frac{D_k}{D} C_{w_k^t}, \\ Q_k & \leftarrow \frac{1}{\text{sim}(C_{w_k^t} \parallel \bar{C}_{w^t})}, \\ C_{\tilde{w}^{t+1}} & \leftarrow \sum_{k \in S_t} \lambda_k C_{w_k^t}, \end{cases} \quad (6)$$

where  $\lambda_k = \frac{Q_k}{\sum_{k \in S_t} Q_k}$  is normalized weight coefficient of the  $k$ -th client,  $\text{sim}$  is cosine similarity used to quantify the degree of deviation between the two classifiers, and  $C_{\tilde{w}^{t+1}}$  indicates the global classifier for the next round. The procedure of *FLPurifier* is presented in Algorithm 1, where the number of clients  $N$  and other parameters is taken as input and the global encoder  $F_{\tilde{w}^T}$  and classifier  $C_{\tilde{w}^T}$  are outputted as the final results.

#### IV. EXPERIMENTAL EVALUATION

To evaluate the effectiveness of our approach, we implemented our *FLPurifier* framework on Pytorch (1.8.0+cu111). All the experiments were conducted on a workstation with windows operating system, Intel i9-9700K CPU, 16GB memory, and NVIDIA GeForce

1) 解耦对比训练: 如前所述, 在本地训练阶段, 每个客户端执行监督对比训练, 该训练由两个阶段组成。具体来说, 在第一阶段, 将在此阶段移除标签的本地数据作为输入, 执行具有非对称 Siamese 网络 (一个在线网络和一个目标网络) 的对比学习。引自 BYOL [46], 在线网络由一个在线编码器和预测器组成, 而目标网络仅包含一个目标编码器。值得注意的是, 与特定分类任务中使用的分类器不同, 预测器用于为在线网络输出对比的回归目标。

在联邦学习 (FL) 中, 除了第一次本地训练外, 我们采用全局编码器  $F_{\tilde{w}^t}$  和预测器  $P_{\tilde{w}^t}$  作为本地在线编码器  $F_{w_k^t}$  和预测器  $P_{w_k^t}$ , 而本地目标编码器由上一轮本地在线编码器  $F_{w_k^{t-1}}$  组成。这种设置有助于分散可能存在于由一个或多个后门编码器聚合的全局编码器中的后门簇。也就是说, 在第一次训练轮次中, 良性客户端将初始全局编码器既作为在线编码器又作为目标编码器, 因此训练的编码器是干净的, 值得信赖。可靠的编码器将在下一轮训练中用作目标编码器, 并净化在线编码器中可能存在的后门。在接下来的联邦通信轮次中, 即使后门编码器再次聚合到全局编码器中, 我们也可以利用上一轮本地编码器来纠正它。形式上, 对比损失可以定义为:

$$\mathcal{L}_{w_k^t, w_k^{t-1}} \triangleq \left\| P_{w_k^t}(z) - F_{w_k^{t-1}}(z') \right\|_2^2 \triangleq 2 - 2 \cdot \frac{\left\langle P_{w_k^t}(z), F_{w_k^{t-1}}(z') \right\rangle}{\|P_{w_k^t}(z)\|_2 \cdot \|F_{w_k^{t-1}}(z')\|_2}, \quad (2)$$

where  $P_{w_k^t}(z)$  是在线预测器的输出,  $F_{w_k^{t-1}}(z')$  表示目标编码器的输出,  $t$  表示联邦学习的通信轮次, 而  $k$  索引第  $k$ -个参与者。在线网络和目标网络以同一图像  $x \sim D$  从  $D_k$  中均匀采样的两个图像增强结果  $z$  和  $z'$  作为输入, 其中在线网络通过梯度下降更新, 目标网络的参数  $w_k^{t-1}$  是每次训练步骤: 后在线网络参数  $w_k^t$  的指数移动平均值

$$w_k^{t-1} \leftarrow mw_k^{t-1} + (1-m)w_k^t, \quad (3)$$

where  $m \in [0, 1]$  是衰减率。

在第一阶段后, 本地客户端通过利用全局特征提取器的信息来训练一个干净的特征提取器。为了进一步实现联邦学习的主要任务并充分利用每个客户端拥有的标签信息, 在第二阶段, 我们将提取器与分类器结合, 并在有标签的数据上以监督学习的方式重新训练整个模型。目标函数如下:

$$w_k^t = \arg \min_{w_k^t} \sum_{(x,y) \in \mathcal{D}_k} \text{CrossEntropy}(C_{w_k^t(x)}(x), y). \quad (4)$$

2) 自适应分类器聚合: 每个客户端在本地训练后, 将本地编码器和分类器发送到服务器进行聚合。然后, *FLPurifier* 分别对编码器进行聚合和分类器进行聚合。对于编码器聚合, 服务器根据本地数据大小对参数进行平均, 以创建全局编码器:

$$F_{\tilde{w}^{t+1}} \leftarrow \sum_{k \in S_t} \frac{D_k}{D} F_{w_k^t}, \quad (5)$$

其中  $S_t$  表示为第  $t$ -轮联邦学习所选客户端的集合,  $F_{\tilde{w}^{t+1}}$  表示下一轮的全局编码器。

对于分类器聚合, 考虑到分类器是通过标记数据来完成的特定分类任务, 而攻击者将训练一个连接后门属性与目标标签的分类器。虽然当提取器与后门分类器结合时, 由于提取器是通过对比学习获得的且是干净的, 后门分类器不会导致误分类, 但后门分类器可能会降低准确率并破坏全局模型的性能。因此, *FLPurifier* 引入了一种自适应分类器聚合策略, 以减轻后门分类器造成的不利影响, 进一步提高主任务的准确率。具体来说, 我们根据公式 6 计算反映所有客户端平均水平的平均分类器  $\bar{C}_{w^t}$ 。然后, 每个分类器的权重系数  $Q_k$  由本地分类器与平均分类器之间的偏差程度决定。更精确地说, 本地分类器与平均值的偏差越大, 其权重系数就越小。通常, 后门分类器与正常分类器存在显著差异, 因此其系数相对较低, 导致其对全局分类器的影响有限。最后, 服务器根据其系数执行自适应分类器聚合:

$$\begin{cases} \bar{C}_{w^t} & \leftarrow \sum_{k \in S_t} \frac{D_k}{D} C_{w_k^t}, \\ Q_k & \leftarrow \frac{1}{\text{sim}(C_{w_k^t} \parallel \bar{C}_{w^t})}, \\ C_{\tilde{w}^{t+1}} & \leftarrow \sum_{k \in S_t} \lambda_k C_{w_k^t}, \end{cases} \quad (6)$$

where  $\lambda_k = \frac{Q_k}{\sum_{k \in S_t} Q_k}$  is normalized weight coefficient of the  $k$ -th client,  $\text{sim}$  is cosine similarity used to quantify the degree of deviation between the two classifiers, and  $C_{\tilde{w}^{t+1}}$  indicates the global classifier for the next round. The procedure of *FLPurifier* is presented in Algorithm 1, where the number of clients  $N$  and other parameters is taken as input and the global encoder  $F_{\tilde{w}^T}$  and classifier  $C_{\tilde{w}^T}$  are outputted as the final results.

#### IV. EXPERIMENTAL EVALUATION

To evaluate the effectiveness of our approach, we implemented our *FLPurifier* framework on Pytorch (1.8.0+cu111). All the experiments were conducted on a workstation with windows operating system, Intel i9-9700K CPU, 16GB memory, and NVIDIA GeForce



**Algorithm 1** *FLPurifier* Algorithm

**Input:** number of clients  $N$ , number of communication round  $T$ , learning rate  $\eta$ , contrastive epoch  $E_{con}$ , supervised epoch  $E_{sup}$ .

**Output:** The final global encoder  $F_{\tilde{w}^T}$  and classifier  $C_{\tilde{w}^T}$ .

- 1 **Server executes:**
- 2 Initialize the global encoder, predictor and classifier  $F_{\tilde{w}^0}, P_{\tilde{w}^0}, C_{\tilde{w}^0}$ ;
- 3 **for each communication round**  $t \in [1, 2, \dots, T]$  **do**
- 4    $S_t \leftarrow$  (random set of  $N$  clients);
- 5   **for each client**  $k \in S_t$  **do**
- 6      $F_{w_k^t}, C_{w_k^t}$
- 7      $\leftarrow$  **LocalTraining**( $k, F_{\tilde{w}^t}, C_{\tilde{w}^t}$ );
- 8     // Encoder, predictor aggregation
- 9      $F_{\tilde{w}^{t+1}} \leftarrow \sum_{k \in S_t} \frac{D_k}{D} F_{w_k^t}$ ;
- 10    // Adaptive classifier aggregation
- 11     $\bar{C}_{w^t} \leftarrow \sum_{k \in S_t} \frac{D_k}{D} C_{w_k^t}$ ;
- 12    **for each client**  $k \in S_t$  **do**
- 13      $Q_k \leftarrow \frac{1}{sim(C_{w_k^t} || \bar{C}_{w^t})}$ ;
- 14     $C_{\tilde{w}^{t+1}} \leftarrow \sum_{k \in S_t} \frac{Q_k}{\sum_{k \in S_t} Q_k} C_{w_k^t}$ ;
- 15 **Return**  $F_{\tilde{w}^T}, G_{\tilde{w}^T}$ ;
- 16 **Client executes:**
- 17 **LocalTraining** ( $k, F_{\tilde{w}^t}, C_{\tilde{w}^t}$ ) ;
- 18  $F_{w_k^t}, C_{w_k^t} \leftarrow F_{\tilde{w}^t}, C_{\tilde{w}^t}$  ;
- 19    // Contrastive training for Extractor
- 20    **for each local epoch**  $k \in [1, 2, \dots, E_{con}]$  **do**
- 21     **for each batch**  $b = \{x\} \in D_k$  **do**
- 22        $w_k^t \leftarrow w_k^t - \eta \nabla \mathcal{L}_{w_k^t, w_k^{t-1}}(w_k^t; x)$ ;
- 23        $w_k^{t-1} \leftarrow m w_k^{t-1} + (1 - m) w_k^t$ ;
- 24     // Supervised training for Classifier
- 25     **for each local epoch**  $k \in [1, 2, \dots, E_{sup}]$  **do**
- 26       **for each batch**  $b = \{x, y\} \in D_k$  **do**
- 27           $\mathcal{L}_{sup} \leftarrow CrossEntropy(C_{w_k^t}(x), y)$ ;
- 28           $w_k^t \leftarrow w_k^t - \eta \nabla \mathcal{L}_{sup}$ ;
- 29 **Return**  $F_{w_k^t}, C_{w_k^t}$ .

GTX2080 GPU. In this section, we designed comprehensive experiments to answer the following three research questions.

- **RQ1** (Superiority of *FLPurifier*): What are the advantages of *FLPurifier* compared with state-of-the-art (SOTA) methods?
- **RQ2** (Effectiveness of *FLPurifier*): How can *FLPurifier* achieve backdoor defense during training phase?
- **RQ3** (Applicability of *FLPurifier*): Does *FLPurifier* still be effective in different FL settings?
- **RQ4** (Overhead of *FLPurifier*): How does the communication and computational cost of *FLPurifier* perform?

**A. Datasets and Experimental Setup**

1) *Datasets*: *FLPurifier* is evaluated on six widely-used benchmark datasets: MNIST, Fashion-MNIST (F-MNIST), EMNIST, CIFAR-10, GTSRB, and CIFAR-100. Note that EMNIST is used for Non-IID settings. For the encoder, *FLPurifier* is compatible with various model architectures. We consider several popular models: CNN model for MNIST and F-MNIST, VGG for CIFAR-10, and DenseNet for GTSRB. For the CIFAR-100 which is a 100-classes dataset, we employ Resnet-18.

a) *Training configurations*: In our experiments, federated training protocol is run with 100 clients (with averaged training samples) where 30 of them are malicious. The attackers embed a trigger in the training images and assign it with a target label and all the attackers construct the backdoor trigger in the same way for a consistent target. During the training procedure, each participant uses SGD as optimizer and trains for  $E$  local epochs with local learning rate  $\eta$ , and a batch size of 64. For the local training process, we set the attacker's local epochs and initial learning rate as  $E = 20$  and  $\eta = 0.05$ , while the benign participants are  $E = 10$  and  $\eta = 0.1$ . Besides, 10 participants (three of which are malicious) are randomly selected in each round to submit local updates for aggregation. Such settings make contribution to the success of backdoor attacks. Note that above settings are used for backdoor attacks based on different types of triggers while we follow original settings for the baseline attacks.

b) *Evaluation metrics*: We evaluate the performance of defense mechanisms with two metrics: attack success rate (ASR), which is the ratio of backdoored samples misclassified as the labels specified by attackers, and the accuracy of the main classification task on normal samples (MA). A robust FL framework means significantly reducing the ASR while maintaining a high MA.

c) *Baselines*: We test the effect of the defense methods against three kinds of known backdoor attacks in FL, i.e., CAS [4], DBA [5], and Edge-case (PGD) [18]. Moreover, we also choose four typical backdoor defense methods in FL as our comparison baselines, i.e., FoolsGold [10], CONTRA [13], Baffle [11], and RLR [14].

**B. Comparison With SOTA Methods (RQ1)**

Table II shows the performance of *FLPurifier* compared with four baseline backdoor defense methods, all these methods using the same settings as in the original works. For backdoor attacks, we set the scaling factor  $S = 30$  in CAS attack, and other attacks follow their original setting. Besides, column “Before” is listed as the original baseline without any defense and the best results are in Bold.

The experimental show that *FLPurifier* can lower the ASR of all four backdoor attacks to 4% on the four benchmark datasets, which surpasses the performance of the other three defense methods. Overall, all defense methods in the experiments are effective against centralized backdoor attacks while FoolsGold fails to defend against DBA which is in a distributed manner. Specifically, CONTRA and Baffle have a similar level in the defense of backdoor attacks while the effect

**算法 1** *FLPurifier* 算法

**输入:** 客户端数量  $N$ , 数量 of iid  $T$  li  
 commun cat on roun , earn ng rate  $\eta$ ,  
 p contrastive 输出 最终全局编码器  $F_{\tilde{w}^0}$  和分类器  $C_{\tilde{w}^0}$ .

**服务端执行:**

- 1 初始化全局编码器、预测器和分类器  $F_{\tilde{w}^0}, P_{\tilde{w}^0}, C_{\tilde{w}^0}$ ;
- 2 **for 每个通信轮次**  $t \in [1, 2, \dots, T]$  **do**
- 3    $S_t \leftarrow$  ( 随机  $N$  客户端 )
- 4   **对于每个客户端**  $k \in S_t$  **做**
- 5      $F_{w_k^t}, C_{w_k^t}$
- 6     **本地训练 k, F, C;**
- 7     // Encoder, predictor aggregation
- 8      $F_{\tilde{w}^{t+1}} \leftarrow \sum_{k \in S_t} \frac{D_k}{D} F_{w_k^t}$ ;
- 9     // 自适应分类器聚合  $C$
- 10     $\bar{C}_{w^t} \leftarrow \sum_{k \in S_t} \frac{D_k}{D} C_{w_k^t}$ ;
- 11    **对于每个客户端**  $k \in S_t$  **做**
- 12      $Q_k \leftarrow \frac{1}{sim(C_{w_k^t} || \bar{C}_{w^t})}$ ;
- 13     $C_{\tilde{w}^{t+1}} \leftarrow \sum_{k \in S_t} \frac{Q_k}{\sum_{k \in S_t} Q_k} C_{w_k^t}$ ;
- 14 **返回**  $F_{\tilde{w}^T}, G_{\tilde{w}^T}$ ;
- 15 **客户端执行:**
- 16 **LocalTraining** ( $k, F_{\tilde{w}^t}, C_{\tilde{w}^t}$ );  $F_{w_k^t}, C_{w_k^t} \leftarrow F_{\tilde{w}^t}, C_{\tilde{w}^t}$  ;
- 17    // 对 Extractor 进行对比训练
- 18    **对于每个本地轮次**  $k \in [1, 2, \dots, E_{con}]$  **进行**
- 19     **对于每个批次**  $b = \{x\} \in D_k$  **进行**
- 20        $w_k^t \leftarrow w_k^t - \eta \nabla \mathcal{L}_{w_k^t, w_k^{t-1}}(w_k^t; x)$ ;
- 21        $w_k^{t-1} \leftarrow m w_k^{t-1} + (1 - m) w_k^t$ ;
- 22     // 分类器的监督训练
- 23     **对于 每个本地轮次**  $k \in [1, 2, \dots, E_{sup}]$  **执行**
- 24       **对于 每个批次**  $b = \{x, y\} \in D_k$  **执行**
- 25           $\mathcal{L}_{sup} \leftarrow$  交叉熵  $(C_{w_k^t}(x), y)$ ;
- 26           $w_k^t \leftarrow w_k^t - \eta \nabla \mathcal{L}_{sup}$ ;
- 27 **返回**  $F_{w_k^t}, C_{w_k^t}$  .

GTX2080 GPU. 在本节中，我们设计了全面的实验来回答以下三个研究问题。

- **RQ1** (Superiority of *FLPurifier*): What are the advantages of *FLPurifier* compared with state-of-the-art (SOTA) methods?
- **RQ2** (Effectiveness of *FLPurifier*): How can *FLPurifier* achieve backdoor defense during training phase?
- **RQ3** (Applicability of *FLPurifier*): Does *FLPurifier* still be effective in different FL settings?
- **RQ4** (Overhead of *FLPurifier*): How does the communication and computational cost of *FLPurifier* perform?

**A. 数据集和实验设置**

1) 数据集: *FLPurifier* 在六个广泛使用的基准数据集上进行了评估: MNIST、Fashion-MNIST (F-MNIST)、EMNIST、CIFAR-10、GTSRB 和 CIFAR-100。请注意, EMNIST 用于非独立同分布设置。对于编码器, *FLPurifier* 兼容各种模型架构。我们考虑了几个流行的模型: 用于 MNIST 和 F-MNIST 的 CNN 模型, 用于 CIFAR-10 的 VGG, 以及用于 GTSRB 的 DenseNet。对于 100 个类别的数据集 CIFAR-100, 我们采用了 Resnet-18。

a) 训练配置: 在我们的实验中, 联邦训练协议在 100 个客户端 (具有平均训练样本) 上运行, 其中 30 个是恶意的。攻击者在训练图像中嵌入触发器, 并将其分配给目标标签, 所有攻击者都以相同的方式构建后门触发器以保持一致的目标。在训练过程中, 每个参与者使用 SGD 作为优化器, 并训练  $E$  个本地轮次, 本地学习率为  $\eta$ , 批大小为 64。对于本地训练过程, 我们将攻击者的本地轮次和初始学习率设置为  $E = 20$  和  $\eta = 0.05$ , 而良性参与者为  $E = 10$  和  $\eta = 0.1$ 。此外, 每轮随机选择 10 个参与者 (其中三个是恶意的) 提交本地更新以进行聚合。这些设置促成了后门攻击的成功。请注意, 上述设置用于基于不同类型触发器的后门攻击, 而我们遵循原始设置用于基线攻击。

b) 评估指标: 我们使用两个指标评估防御机制的性能: 攻击成功率 (ASR), 即被攻击者指定的标签错误分类的后门样本的比例, 以及正常样本的主要分类任务准确率 (MA)。一个鲁棒的联邦学习框架意味着显著降低 ASR, 同时保持高 MA。

c) 基线: 我们测试了防御方法对联邦学习中三种已知后门攻击的效果, 即 CAS [4], DBA [5], 和边缘情况 (PGD) [18]。此外, 我们还选择了联邦学习中四种典型的后门防御方法作为我们的比较基线, 即 FoolsGold [10], CONTRA [13], Baffle [11], 和 RLR [14]。

**B. 与 SOTA 方法的比较 (RQ1)**

表 II 展示了 *FLPurifier* 与四种基线后门防御方法的性能比较, 所有这些方法都使用原始工作中的相同设置。对于后门攻击, 我们设置了 CAS 攻击中的缩放因子  $S = 30$ , 其他攻击遵循其原始设置。此外, “Before” 列列出了没有任何防御的原始基线, 最佳结果以粗体显示。实验表明, *FLPurifier* 可以将所有四种后门攻击在四个基准数据集上的 ASR 降低至 4%, 这超过了其他三种防御方法的性能。总体而言, 实验中的所有防御方法对集中式后门攻击都有效, 而 FoolsGold 无法防御分布式方式的后门攻击 DBA。具体来说, CONTRA 和 Baffle 在后门攻击防御方面具有相似的水平, 而效果



TABLE II  
PERFORMANCE COMPARISON BETWEEN *FLPurifier* AND FOUR SOTA DEFENSE METHODS

| Datasets  | Attack    | Before |              | FoolsGold |       | CONTRA |       | Baffle |       | RLR  |       | <i>FLPurifier</i> |              |
|-----------|-----------|--------|--------------|-----------|-------|--------|-------|--------|-------|------|-------|-------------------|--------------|
|           |           | ASR    | MA           | ASR       | MA    | ASR    | MA    | ASR    | MA    | ASR  | MA    | ASR               | MA           |
| MNIST     | DBA       | 100    | <b>90.43</b> | 97.73     | 79.23 | 7.13   | 87.45 | 8.64   | 82.97 | 6.51 | 86.19 | <b>2.69</b>       | 90.32        |
|           | Edge-case | 92.47  | <b>93.01</b> | 12.59     | 81.51 | 5.34   | 88.13 | 7.43   | 84.06 | 2.37 | 88.52 | <b>1.47</b>       | 92.57        |
|           | PGD       | 90.39  | 91.12        | 11.87     | 82.05 | 4.29   | 88.26 | 3.26   | 85.72 | 1.94 | 85.88 | <b>0.94</b>       | <b>92.05</b> |
| F-MNIST   | DBA       | 100    | 89.67        | 98.69     | 80.05 | 6.79   | 86.92 | 9.05   | 84.39 | 5.64 | 84.47 | <b>2.31</b>       | <b>90.19</b> |
|           | Edge-case | 91.58  | <b>92.56</b> | 10.52     | 81.20 | 2.85   | 87.48 | 3.47   | 85.83 | 3.05 | 85.93 | <b>0.85</b>       | 91.44        |
|           | PGD       | 89.76  | <b>92.03</b> | 11.62     | 79.28 | 3.27   | 88.03 | 4.71   | 83.85 | 2.12 | 87.04 | <b>1.26</b>       | 91.89        |
| CIFAR-10  | CAS       | 79.24  | 87.21        | 9.15      | 80.16 | 3.44   | 82.79 | 3.58   | 79.41 | 8.85 | 82.58 | <b>1.63</b>       | <b>88.14</b> |
|           | DBA       | 94.12  | 72.14        | 92.42     | 71.57 | 8.51   | 71.55 | 9.82   | 70.56 | 7.92 | 69.87 | <b>3.24</b>       | <b>75.66</b> |
|           | Edge-case | 58.61  | <b>83.56</b> | 8.31      | 81.89 | 4.39   | 80.97 | 5.95   | 79.13 | 5.91 | 80.92 | <b>1.32</b>       | 82.43        |
|           | PGD       | 68.35  | <b>79.43</b> | 7.94      | 78.50 | 5.36   | 76.19 | 6.70   | 77.32 | 5.14 | 73.65 | <b>1.58</b>       | 79.21        |
| GTSRB     | CS        | 81.77  | <b>89.53</b> | 10.58     | 80.21 | 5.98   | 87.20 | 7.93   | 83.39 | 5.06 | 85.78 | <b>2.17</b>       | 89.11        |
|           | DBA       | 100    | 89.37        | 98.95     | 79.61 | 6.80   | 86.82 | 7.11   | 82.35 | 5.87 | 87.85 | <b>2.50</b>       | <b>89.92</b> |
|           | Edge-Case | 87.32  | <b>92.59</b> | 10.64     | 81.40 | 3.16   | 85.30 | 4.56   | 84.44 | 1.78 | 86.28 | <b>1.79</b>       | 90.87        |
|           | PGD       | 90.26  | <b>91.93</b> | 11.61     | 79.76 | 3.69   | 86.10 | 4.77   | 82.20 | 1.45 | 85.84 | <b>1.56</b>       | 89.27        |
| CIFAR-100 | CS        | 76.32  | 61.36        | 8.84      | 55.31 | 3.10   | 56.41 | 3.97   | 54.85 | 3.49 | 58.09 | <b>2.07</b>       | <b>62.02</b> |
|           | DBA       | 86.91  | 61.95        | 92.54     | 54.33 | 8.38   | 57.15 | 7.55   | 53.58 | 5.86 | 56.68 | <b>2.81</b>       | <b>62.36</b> |
|           | Edge-Case | 53.25  | <b>63.01</b> | 8.35      | 56.58 | 4.02   | 55.45 | 5.26   | 53.44 | 5.49 | 55.11 | <b>0.96</b>       | 61.65        |
|           | PGD       | 59.17  | <b>61.65</b> | 8.14      | 57.00 | 5.39   | 55.96 | 3.94   | 54.63 | 3.69 | 57.90 | <b>1.41</b>       | 60.18        |

of FoolsGold is slightly inferior. As the backdoor mitigation method, RLR relieving backdoor with robust learning rate shows better performance in defending backdoor against in FL. However, all of the baseline defense methods cause degeneration in ACC while *FLPurifier* exhibits a superior defense effect and maintains model performance degradation at 2%, which is a distinct superiority of *FLPurifier*. Surprisingly, some values of MA from *FLPurifier* are even higher than the “Before” column. To our analysis, it is thanks to the adaptive classifier aggregation that can also reduce the weight of low-quality classifiers.

### C. Further Understanding of *FLPurifier* (RQ2)

1) *Robustness Under Different Triggers*: To further verify the robustness of *FLPurifier* To further verify the robustness of *FLPurifier* to different trigger types and sizes, we select four types of triggers: watermarking, pixel blocks, random noise, and imperceptible input. For the watermarking attack, 0 is applied as the watermarking and the coefficient is varied from 0.1 to 0.5, where their span is 0.2. In the backdoor attacks based on pixel blocks, we extend the trigger sizes from  $3 \times 3$  to  $5 \times 5$  and  $7 \times 7$ . Besides, for the random noise attacks, we change the noise tensity from 10 to 20 and 30. All aforementioned backdoor triggers used in our experiments are shown in Fig. 3.

Table III, IV, and V summarize the performance comparison of *FLPurifier* with other backdoor defense baselines. Overall, all defense methods are nearly immune to the three trigger types and sizes above. In more detail, Baffle detects abnormal updates and CONTRA adjusts the weight of malicious clients. For RLR, it relieves the backdoor with a robust learning rate. The bigger sizes and higher intensity seem to improve the ASR without any backdoor defense. It is because the feature of the trigger is harder with the increase of the trigger sizes,

which leads to the feature being easier to be learned by the model. For different types of triggers, the ASR of the imperceptible input is significantly greater than the other three types of triggers. However, after the defenses are deployed, the backdoored updates are hard to be aggregated into the global model, or their weight is lowered, no matter the trigger types and sizes. For the *FLPurifier*, we directly cut off the connection between trigger features and the target label in the training process, which is also independent of the trigger type. The above findings also indicate that the trigger size and type, which are significant factors in centralized backdoor research, hardly affect the performance of defenses in FL.

In addition, we partly visualize and compare the confusion matrix without and with *FLPurifier* in Fig. 4 under the attack whose triggers are pixel blocks. In MNIST task, the test samples in which the original labels are “0” were embedded in a  $4 \times 4$  pattern and assigned with a target label “8”. For CIFAR-10 task, we respectively assign the 4-th class (“deer”) with the label of the 7-th class (“horse”) under the same setting as MNIST. Intuitively, results indicate that most of the samples “7” with triggers are misclassified into “4”. The same situation occurs in CIFAR-10 dataset. However, the vast majority of crafted inputs are classified into their original labels with the help of *FLPurifier*.

Moreover, we visualize the distribution in feature space of the global extractor under the above attacks with/without *FLPurifier* to further understand the advantages of our proposed method. More specifically, we poisoned 10% of the test data in CIFAR-10, where the number of each poisoned class is the same. Then, the poisoned test data are used for t-SNE visualization. Fig. 5 including six subfigures shows the t-SNE vitalization. From Fig. 5a to Fig. 5b, we can see that the feature representations of benign samples from the same category form an individual cluster, while the poisoned samples form a new cluster (in black color). However, the clusters of

表 II *FLPurifier* 与四种 SOTA 防御方法的性能比较

| Datasets  | Attack    | Before |              | FoolsGold |       | CONTRA |       | Baffle |       | RLR  |       | <i>FLPurifier</i> |              |
|-----------|-----------|--------|--------------|-----------|-------|--------|-------|--------|-------|------|-------|-------------------|--------------|
|           |           | ASR    | MA           | ASR       | MA    | ASR    | MA    | ASR    | MA    | ASR  | MA    | ASR               | MA           |
| MNIST     | DBA       | 100    | <b>90.43</b> | 97.73     | 79.23 | 7.13   | 87.45 | 8.64   | 82.97 | 6.51 | 86.19 | <b>2.69</b>       | 90.32        |
|           | Edge-case | 92.47  | <b>93.01</b> | 12.59     | 81.51 | 5.34   | 88.13 | 7.43   | 84.06 | 2.37 | 88.52 | <b>1.47</b>       | 92.57        |
|           | PGD       | 90.39  | 91.12        | 11.87     | 82.05 | 4.29   | 88.26 | 3.26   | 85.72 | 1.94 | 85.88 | <b>0.94</b>       | <b>92.05</b> |
| F-MNIST   | DBA       | 100    | 89.67        | 98.69     | 80.05 | 6.79   | 86.92 | 9.05   | 84.39 | 5.64 | 84.47 | <b>2.31</b>       | <b>90.19</b> |
|           | Edge-case | 91.58  | <b>92.56</b> | 10.52     | 81.20 | 2.85   | 87.48 | 3.47   | 85.83 | 3.05 | 85.93 | <b>0.85</b>       | 91.44        |
|           | PGD       | 89.76  | <b>92.03</b> | 11.62     | 79.28 | 3.27   | 88.03 | 4.71   | 83.85 | 2.12 | 87.04 | <b>1.26</b>       | 91.89        |
| CIFAR-10  | CAS       | 79.24  | 87.21        | 9.15      | 80.16 | 3.44   | 82.79 | 3.58   | 79.41 | 8.85 | 82.58 | <b>1.63</b>       | <b>88.14</b> |
|           | DBA       | 94.12  | 72.14        | 92.42     | 71.57 | 8.51   | 71.55 | 9.82   | 70.56 | 7.92 | 69.87 | <b>3.24</b>       | <b>75.66</b> |
|           | Edge-case | 58.61  | <b>83.56</b> | 8.31      | 81.89 | 4.39   | 80.97 | 5.95   | 79.13 | 5.91 | 80.92 | <b>1.32</b>       | 82.43        |
|           | PGD       | 68.35  | <b>79.43</b> | 7.94      | 78.50 | 5.36   | 76.19 | 6.70   | 77.32 | 5.14 | 73.65 | <b>1.58</b>       | 79.21        |
| GTSRB     | CS        | 81.77  | <b>89.53</b> | 10.58     | 80.21 | 5.98   | 87.20 | 7.93   | 83.39 | 5.06 | 85.78 | <b>2.17</b>       | 89.11        |
|           | DBA       | 100    | 89.37        | 98.95     | 79.61 | 6.80   | 86.82 | 7.11   | 82.35 | 5.87 | 87.85 | <b>2.50</b>       | <b>89.92</b> |
|           | Edge-Case | 87.32  | <b>92.59</b> | 10.64     | 81.40 | 3.16   | 85.30 | 4.56   | 84.44 | 1.78 | 86.28 | <b>1.79</b>       | 90.87        |
|           | PGD       | 90.26  | <b>91.93</b> | 11.61     | 79.76 | 3.69   | 86.10 | 4.77   | 82.20 | 1.45 | 85.84 | <b>1.56</b>       | 89.27        |
| CIFAR-100 | CS        | 76.32  | 61.36        | 8.84      | 55.31 | 3.10   | 56.41 | 3.97   | 54.85 | 3.49 | 58.09 | <b>2.07</b>       | <b>62.02</b> |
|           | DBA       | 86.91  | 61.95        | 92.54     | 54.33 | 8.38   | 57.15 | 7.55   | 53.58 | 5.86 | 56.68 | <b>2.81</b>       | <b>62.36</b> |
|           | Edge-Case | 53.25  | <b>63.01</b> | 8.35      | 56.58 | 4.02   | 55.45 | 5.26   | 53.44 | 5.49 | 55.11 | <b>0.96</b>       | 61.65        |
|           | PGD       | 59.17  | <b>61.65</b> | 8.14      | 57.00 | 5.39   | 55.96 | 3.94   | 54.63 | 3.69 | 57.90 | <b>1.41</b>       | 60.18        |

FoolsGold 的性能略差。作为后门缓解方法, RLR 通过鲁棒学习率缓解后门, 在 FL 中防御后门时表现更好。然而, 所有基线防御方法都会导致 ACC 退化, 而 *FLPurifier* 表现出更优越的防御效果, 并使模型性能退化保持在 2%, 这是 *FLPurifier* 的独特优势。令人惊讶的是, *FLPurifier* 的一些 MA 值甚至高于 “之前” 列。据我们的分析, 这要归功于自适应分类器聚合, 它也能减少低质量分类器的权重。

### C. 进一步理解 *FLPurifier* (RQ2)

1) 不同触发器下的鲁棒性: 为了进一步验证 *FLPurifier* 为了进一步验证 *FLPurifier* 对不同触发器类型和大小的鲁棒性, 我们选择了四种触发器: 水印、像素块、随机噪声和不可感知输入。在水印攻击中, 0 被用作水印, 系数从 0.1 变化到 0.5, 其跨度为 0.2。在基于像素块的后门攻击中, 我们将触发器大小从  $3 \times 3$  扩展到  $5 \times 5$  和  $7 \times 7$ 。此外, 对于随机噪声攻击, 我们将噪声强度从 10 改为 20 和 30。我们实验中使用的所有上述后门触发器如图 3 所示。

Table III, IV, and V summarize the performance comparison of *FLPurifier* with other backdoor defense baselines. Overall, all defense methods are nearly immune to the three trigger types and sizes above. In more detail, Baffle detects abnormal updates and CONTRA adjusts the weight of malicious clients. For RLR, it relieves the backdoor with a robust learning rate. The bigger sizes and higher intensity seem to improve the ASR without any backdoor defense. It is because the feature of the trigger is harder with the increase of the trigger sizes,

which leads to the feature being easier to be learned by the model. For different types of triggers, the ASR of the imperceptible input is significantly greater than the other three types of triggers. However, after the defenses are deployed, the backdoored updates are hard to be aggregated into the global model, or their weight is lowered, no matter the trigger types and sizes. For the *FLPurifier*, we directly cut off the connection between trigger features and the target label in the training process, which is also independent of the trigger type. The above findings also indicate that the trigger size and type, which are significant factors in centralized backdoor research, hardly affect the performance of defenses in FL.

此外, 我们部分可视化和比较了在像素块触发攻击下, 没有和有 *FLPurifier* 的混淆矩阵, 如图 4 所示。在 MNIST 任务中, 原始标签为 “0” 的测试样本被嵌入到  $4 \times 4$  模式中, 并被分配目标标签 “8”。对于 CIFAR-10 任务, 我们在与 MNIST 相同的设置下, 分别将第 4 类 (“鹿”) 分配为第 7 类 (“马”) 的标签。直观上, 结果表明, 大多数带有触发器的样本 “7” 被错误分类为 “4”。在 CIFAR-10 数据集上也出现了相同的情况。然而, 在 *FLPurifier* 的帮助下, 绝大多数构造的输入被分类为其原始标签。

此外, 我们可视化了在上述攻击下, 全局提取器在有 *FLPurifier* 的特征空间分布, 以进一步理解我们提出的方法的优势。更具体地说, 我们在 CIFAR-10 中污染了 10% 的测试数据, 其中每个污染类别的数量相同。然后, 使用污染的测试数据进行 t-SNE 可视化。图 5 包括六个子图, 展示了 t-SNE 的激活效果。从图 5a 到图 5b, 我们可以看到, 来自同一类别的良性样本的特征表示形成一个独立的簇, 而污染样本形成一个新的簇 (黑色)。然而,



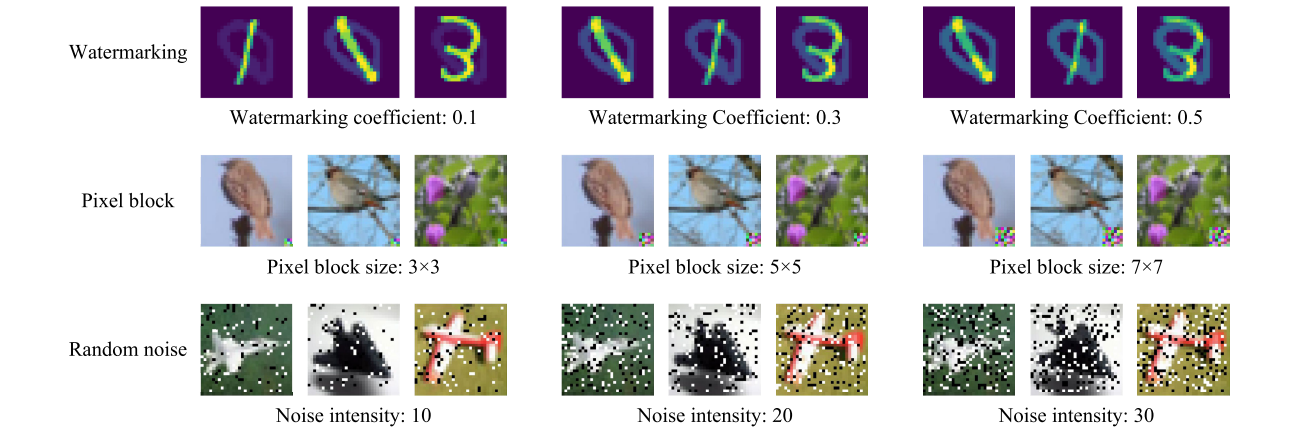


Fig. 3. Examples of backdoored images by different trigger types and size.

TABLE III  
PERFORMANCE OF *FLPurifier* UNDER ATTACKS WITH DIFFERENT TYPES OF TRIGGERS ON MNIST

| Backdoor attacks    | Trigger size | Before |              | FoolsGold |       | CONTRA |       | Baffle |       | RLR  |       | <i>FLPurifier</i> |              |
|---------------------|--------------|--------|--------------|-----------|-------|--------|-------|--------|-------|------|-------|-------------------|--------------|
|                     |              | ASR    | MA           | ASR       | MA    | ASR    | MA    | ASR    | MA    | ASR  | MA    | ASR               | MA           |
| Imperceptible input |              | 93.78  | <b>91.32</b> | 18.87     | 81.35 | 13.42  | 84.27 | 15.92  | 81.41 | 6.62 | 86.42 | <b>2.92</b>       | 90.55        |
| Watermarking        | 0.1          | 89.06  | <b>91.58</b> | 11.02     | 82.23 | 4.02   | 87.45 | 5.38   | 85.43 | 3.03 | 88.24 | <b>1.24</b>       | 91.41        |
|                     | 0.3          | 91.25  | 92.83        | 12.75     | 81.51 | 3.68   | 88.13 | 4.75   | 84.35 | 2.84 | 88.01 | <b>1.49</b>       | <b>93.01</b> |
|                     | 0.5          | 92.87  | <b>92.88</b> | 10.72     | 82.05 | 4.36   | 88.26 | 4.86   | 85.65 | 2.59 | 87.79 | <b>2.05</b>       | 90.88        |
| Pixel block         | 3x3          | 90.44  | <b>90.89</b> | 10.44     | 82.05 | 3.52   | 86.92 | 3.74   | 84.51 | 3.12 | 88.25 | <b>0.89</b>       | 89.96        |
|                     | 5x5          | 92.18  | <b>93.54</b> | 9.32      | 81.20 | 3.44   | 87.48 | 3.92   | 86.51 | 3.13 | 88.57 | <b>1.35</b>       | 93.20        |
|                     | 7x7          | 90.11  | <b>90.89</b> | 10.02     | 82.28 | 2.12   | 88.03 | 3.73   | 85.90 | 1.66 | 87.32 | <b>1.26</b>       | 90.17        |
| Random noise        | 10           | 89.48  | <b>92.60</b> | 11.43     | 81.16 | 3.73   | 87.79 | 4.19   | 85.50 | 2.57 | 86.71 | <b>1.63</b>       | 89.60        |
|                     | 20           | 90.01  | <b>93.29</b> | 14.93     | 82.57 | 4.10   | 87.55 | 4.53   | 85.49 | 3.44 | 87.89 | <b>2.24</b>       | 91.23        |
|                     | 30           | 91.20  | <b>92.79</b> | 12.25     | 81.89 | 2.94   | 86.97 | 4.78   | 85.20 | 3.39 | 87.62 | <b>2.32</b>       | 91.77        |

TABLE IV  
PERFORMANCE OF *FLPurifier* UNDER ATTACKS WITH DIFFERENT TYPES OF TRIGGERS ON CIFAR-10

| Backdoor attacks    | Trigger size | Before |              | FoolsGold |       | CONTRA |       | Baffle |       | RLR  |       | <i>FLPurifier</i> |              |
|---------------------|--------------|--------|--------------|-----------|-------|--------|-------|--------|-------|------|-------|-------------------|--------------|
|                     |              | ASR    | MA           | ASR       | MA    | ASR    | MA    | ASR    | MA    | ASR  | MA    | ASR               | MA           |
| Imperceptible input |              | 92.79  | <b>86.83</b> | 18.64     | 79.32 | 14.20  | 79.55 | 16.71  | 80.46 | 5.43 | 82.19 | <b>2.92</b>       | 85.97        |
| Watermarking        | 0.1          | 78.53  | 84.43        | 7.82      | 79.65 | 3.57   | 82.04 | 4.06   | 81.97 | 2.74 | 81.59 | <b>1.25</b>       | <b>84.56</b> |
|                     | 0.3          | 79.76  | <b>83.59</b> | 8.63      | 79.32 | 3.70   | 81.35 | 5.62   | 81.06 | 3.36 | 82.52 | <b>1.09</b>       | 83.25        |
|                     | 0.5          | 78.86  | <b>84.35</b> | 6.82      | 80.03 | 3.96   | 82.21 | 4.82   | 80.72 | 3.34 | 83.01 | <b>1.36</b>       | 83.98        |
| Pixel block         | 3x3          | 76.54  | <b>85.42</b> | 8.17      | 79.45 | 2.90   | 82.44 | 3.11   | 79.39 | 2.55 | 82.42 | <b>0.97</b>       | 83.06        |
|                     | 5x5          | 76.19  | 83.23        | 7.71      | 78.94 | 3.21   | 83.17 | 4.06   | 81.83 | 2.79 | 83.04 | <b>1.26</b>       | <b>83.54</b> |
|                     | 7x7          | 76.55  | <b>84.43</b> | 9.77      | 78.95 | 2.53   | 82.19 | 3.86   | 80.85 | 2.74 | 82.14 | <b>0.74</b>       | 82.71        |
| Random noise        | 10           | 77.50  | <b>87.16</b> | 7.14      | 78.62 | 3.42   | 83.75 | 4.68   | 82.41 | 2.77 | 82.47 | <b>1.31</b>       | 84.04        |
|                     | 20           | 76.85  | <b>85.85</b> | 8.67      | 79.37 | 4.18   | 82.61 | 6.70   | 81.56 | 3.35 | 81.60 | <b>2.39</b>       | 85.11        |
|                     | 30           | 76.23  | <b>84.09</b> | 7.51      | 79.18 | 4.68   | 83.13 | 5.67   | 82.13 | 3.99 | 82.51 | <b>1.48</b>       | 83.54        |

poisoned samples in Fig. 5e to Fig. 5f are absolutely damaged, all the poisoned samples afresh assemble with the benign samples from the same category. This means that *FLPurifier* can effectively break the cluster of backdoor features and further prevent the establishment of the connection between the targeted label and the trigger through decoupled contrastive training. Moreover, the benign samples locate in their own cluster in a relatively concentrated manner, where the distance between samples within clusters is very close meanwhile there is a significant distance between clusters, illustrating that the extractor shows excellent ability on the main classify task.

2) *Adaptive Classifier Aggregation*: To further explore the impact of adaptive classifier aggregation (ACA), we process more experiments with/without ACA on IID and Non-IID settings, respectively. For the scene without ACA, we adopt Fedavg algorithm to aggregate the classifiers. Table VI and VII represent the result of the experience. In general, both the defense effect and the main task accuracy have a degeneration after we remove the ACA. In specific, the MA decreased by about 6% at most, which means that the ACA makes significant contributions to the maintenance of the MA. According to our analysis, it is because ACA first calculates the averaged

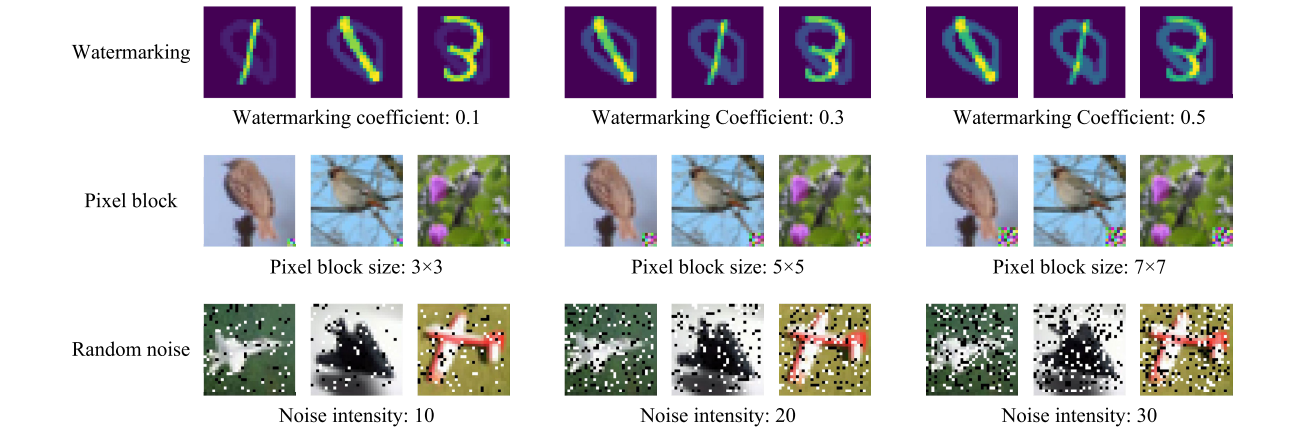


图 3. 不同触发类型和大小后门图像示例。

表 III: MNIST 上不同类型触发器攻击下 *FLPurifier* 的性能

| Backdoor attacks    | Trigger size | Before |              | FoolsGold |       | CONTRA |       | Baffle |       | RLR  |       | <i>FLPurifier</i> |              |
|---------------------|--------------|--------|--------------|-----------|-------|--------|-------|--------|-------|------|-------|-------------------|--------------|
|                     |              | ASR    | MA           | ASR       | MA    | ASR    | MA    | ASR    | MA    | ASR  | MA    | ASR               | MA           |
| Imperceptible input |              | 93.78  | <b>91.32</b> | 18.87     | 81.35 | 13.42  | 84.27 | 15.92  | 81.41 | 6.62 | 86.42 | <b>2.92</b>       | 90.55        |
| Watermarking        | 0.1          | 89.06  | <b>91.58</b> | 11.02     | 82.23 | 4.02   | 87.45 | 5.38   | 85.43 | 3.03 | 88.24 | <b>1.24</b>       | 91.41        |
|                     | 0.3          | 91.25  | 92.83        | 12.75     | 81.51 | 3.68   | 88.13 | 4.75   | 84.35 | 2.84 | 88.01 | <b>1.49</b>       | <b>93.01</b> |
|                     | 0.5          | 92.87  | <b>92.88</b> | 10.72     | 82.05 | 4.36   | 88.26 | 4.86   | 85.65 | 2.59 | 87.79 | <b>2.05</b>       | 90.88        |
| Pixel block         | 3x3          | 90.44  | <b>90.89</b> | 10.44     | 82.05 | 3.52   | 86.92 | 3.74   | 84.51 | 3.12 | 88.25 | <b>0.89</b>       | 89.96        |
|                     | 5x5          | 92.18  | <b>93.54</b> | 9.32      | 81.20 | 3.44   | 87.48 | 3.92   | 86.51 | 3.13 | 88.57 | <b>1.35</b>       | 93.20        |
|                     | 7x7          | 90.11  | <b>90.89</b> | 10.02     | 82.28 | 2.12   | 88.03 | 3.73   | 85.90 | 1.66 | 87.32 | <b>1.26</b>       | 90.17        |
| Random noise        | 10           | 89.48  | <b>92.60</b> | 11.43     | 81.16 | 3.73   | 87.79 | 4.19   | 85.50 | 2.57 | 86.71 | <b>1.63</b>       | 89.60        |
|                     | 20           | 90.01  | <b>93.29</b> | 14.93     | 82.57 | 4.10   | 87.55 | 4.53   | 85.49 | 3.44 | 87.89 | <b>2.24</b>       | 91.23        |
|                     | 30           | 91.20  | <b>92.79</b> | 12.25     | 81.89 | 2.94   | 86.97 | 4.78   | 85.20 | 3.39 | 87.62 | <b>2.32</b>       | 91.77        |

表 IV: CIFAR-10 上不同类型触发器攻击下 *FLPurifier* 的性能

| Backdoor attacks    | Trigger size | Before |              | FoolsGold |       | CONTRA |       | Baffle |       | RLR  |       | <i>FLPurifier</i> |              |
|---------------------|--------------|--------|--------------|-----------|-------|--------|-------|--------|-------|------|-------|-------------------|--------------|
|                     |              | ASR    | MA           | ASR       | MA    | ASR    | MA    | ASR    | MA    | ASR  | MA    | ASR               | MA           |
| Imperceptible input |              | 92.79  | <b>86.83</b> | 18.64     | 79.32 | 14.20  | 79.55 | 16.71  | 80.46 | 5.43 | 82.19 | <b>2.92</b>       | 85.97        |
| Watermarking        | 0.1          | 78.53  | 84.43        | 7.82      | 79.65 | 3.57   | 82.04 | 4.06   | 81.97 | 2.74 | 81.59 | <b>1.25</b>       | <b>84.56</b> |
|                     | 0.3          | 79.76  | <b>83.59</b> | 8.63      | 79.32 | 3.70   | 81.35 | 5.62   | 81.06 | 3.36 | 82.52 | <b>1.09</b>       | 83.25        |
|                     | 0.5          | 78.86  | <b>84.35</b> | 6.82      | 80.03 | 3.96   | 82.21 | 4.82   | 80.72 | 3.34 | 83.01 | <b>1.36</b>       | 83.98        |
| Pixel block         | 3x3          | 76.54  | <b>85.42</b> | 8.17      | 79.45 | 2.90   | 82.44 | 3.11   | 79.39 | 2.55 | 82.42 | <b>0.97</b>       | 83.06        |
|                     | 5x5          | 76.19  | 83.23        | 7.71      | 78.94 | 3.21   | 83.17 | 4.06   | 81.83 | 2.79 | 83.04 | <b>1.26</b>       | <b>83.54</b> |
|                     | 7x7          | 76.55  | <b>84.43</b> | 9.77      | 78.95 | 2.53   | 82.19 | 3.86   | 80.85 | 2.74 | 82.14 | <b>0.74</b>       | 82.71        |
| Random noise        | 10           | 77.50  | <b>87.16</b> | 7.14      | 78.62 | 3.42   | 83.75 | 4.68   | 82.41 | 2.77 | 82.47 | <b>1.31</b>       | 84.04        |
|                     | 20           | 76.85  | <b>85.85</b> | 8.67      | 79.37 | 4.18   | 82.61 | 6.70   | 81.56 | 3.35 | 81.60 | <b>2.39</b>       | 85.11        |
|                     | 30           | 76.23  | <b>84.09</b> | 7.51      | 79.18 | 4.68   | 83.13 | 5.67   | 82.13 | 3.99 | 82.51 | <b>1.48</b>       | 83.54        |

图 5 中的中毒样本 e 到 f 是完全损坏的, 所有中毒样本与同一类别的良性样本重新组合。这意味着 *FLPurifier* 可以通过解耦对比训练有效地打破后门特征的簇, 并进一步防止目标标签与触发器之间的连接。此外, 良性样本在其自己的簇中相对集中地定位, 其中簇内样本之间的距离非常近, 而簇之间的距离显著, 这表明提取器在主要分类任务上表现出色。

2) 自适应分类器聚合: 为了进一步探索自适应分类器聚合 (ACA) 的影响, 我们在 IID 和非 IID 设置下分别处理了有 / 无 ACA 的更多实验。对于没有 ACA 的场景, 我们采用 Fedavg 算法来聚合分类器。表 VI 和 VII 展示了实验结果。一般来说, 在移除 ACA 后, 防御效果和主要任务精度都有退化。具体来说, MA 最多下降了约 6%, 这意味着 ACA 对 MA 的维持做出了显著贡献。根据我们的分析, 这是因为 ACA 首先计算了平均



TABLE V  
PERFORMANCE OF *FLPurifier* UNDER ATTACKS WITH DIFFERENT TYPES OF TRIGGERS ON CIFAR-100

| Backdoor attacks    | Trigger size | Before |              | FoolsGold |       | CONTRA |       | Baffle |       | RLR  |       | FLPurifier  |              |
|---------------------|--------------|--------|--------------|-----------|-------|--------|-------|--------|-------|------|-------|-------------|--------------|
|                     |              | ASR    | MA           | ASR       | MA    | ASR    | MA    | ASR    | MA    | ASR  | MA    | ASR         | MA           |
| Imperceptible input |              | 89.40  | <b>61.27</b> | 30.88     | 51.16 | 7.50   | 57.79 | 9.29   | 55.41 | 2.49 | 57.58 | <b>1.33</b> | 60.65        |
| Watermarking        | 0.1          | 77.07  | <b>62.30</b> | 9.59      | 51.23 | 3.08   | 57.45 | 4.55   | 55.74 | 2.82 | 57.19 | <b>2.31</b> | 61.67        |
|                     | 0.3          | 78.30  | <b>63.54</b> | 11.89     | 51.51 | 4.32   | 57.13 | 4.59   | 55.18 | 3.32 | 56.52 | <b>2.47</b> | 62.45        |
|                     | 0.5          | 79.42  | 62.60        | 10.95     | 52.05 | 3.68   | 57.26 | 3.62   | 55.61 | 2.93 | 57.88 | <b>1.94</b> | <b>62.71</b> |
| Pixel block         | 3x3          | 73.95  | <b>62.19</b> | 9.07      | 52.05 | 3.82   | 58.92 | 2.07   | 57.28 | 2.28 | 58.47 | <b>0.93</b> | 62.12        |
|                     | 5x5          | 75.72  | <b>62.45</b> | 9.79      | 52.20 | 3.11   | 57.48 | 3.53   | 57.01 | 2.27 | 57.93 | <b>0.96</b> | 62.02        |
|                     | 7x7          | 76.48  | <b>61.12</b> | 8.80      | 53.28 | 3.66   | 57.03 | 3.99   | 57.80 | 1.93 | 57.04 | <b>0.76</b> | 60.17        |
| Random noise        | 10           | 79.40  | <b>61.27</b> | 11.88     | 51.16 | 3.50   | 57.79 | 4.29   | 55.39 | 2.49 | 57.58 | <b>1.33</b> | 60.65        |
|                     | 30           | 80.48  | 62.07        | 13.18     | 50.57 | 4.52   | 58.55 | 5.48   | 54.56 | 3.27 | 56.87 | <b>2.24</b> | <b>62.12</b> |
|                     | 50           | 81.92  | <b>61.32</b> | 11.60     | 51.89 | 4.56   | 56.97 | 4.34   | 53.15 | 3.33 | 58.02 | <b>2.37</b> | 59.97        |

表 V 在 CIFAR-100 上遭受不同类型触发器攻击时 *FLPurifier* 的性能

| Backdoor attacks    | Trigger size | Before |              | FoolsGold |       | CONTRA |       | Baffle |       | RLR  |       | FLPurifier  |              |
|---------------------|--------------|--------|--------------|-----------|-------|--------|-------|--------|-------|------|-------|-------------|--------------|
|                     |              | ASR    | MA           | ASR       | MA    | ASR    | MA    | ASR    | MA    | ASR  | MA    | ASR         | MA           |
| Imperceptible input |              | 89.40  | <b>61.27</b> | 30.88     | 51.16 | 7.50   | 57.79 | 9.29   | 55.41 | 2.49 | 57.58 | <b>1.33</b> | 60.65        |
| Watermarking        | 0.1          | 77.07  | <b>62.30</b> | 9.59      | 51.23 | 3.08   | 57.45 | 4.55   | 55.74 | 2.82 | 57.19 | <b>2.31</b> | 61.67        |
|                     | 0.3          | 78.30  | <b>63.54</b> | 11.89     | 51.51 | 4.32   | 57.13 | 4.59   | 55.18 | 3.32 | 56.52 | <b>2.47</b> | 62.45        |
|                     | 0.5          | 79.42  | 62.60        | 10.95     | 52.05 | 3.68   | 57.26 | 3.62   | 55.61 | 2.93 | 57.88 | <b>1.94</b> | <b>62.71</b> |
| Pixel block         | 3x3          | 73.95  | <b>62.19</b> | 9.07      | 52.05 | 3.82   | 58.92 | 2.07   | 57.28 | 2.28 | 58.47 | <b>0.93</b> | 62.12        |
|                     | 5x5          | 75.72  | <b>62.45</b> | 9.79      | 52.20 | 3.11   | 57.48 | 3.53   | 57.01 | 2.27 | 57.93 | <b>0.96</b> | 62.02        |
|                     | 7x7          | 76.48  | <b>61.12</b> | 8.80      | 53.28 | 3.66   | 57.03 | 3.99   | 57.80 | 1.93 | 57.04 | <b>0.76</b> | 60.17        |
| Random noise        | 10           | 79.40  | <b>61.27</b> | 11.88     | 51.16 | 3.50   | 57.79 | 4.29   | 55.39 | 2.49 | 57.58 | <b>1.33</b> | 60.65        |
|                     | 30           | 80.48  | 62.07        | 13.18     | 50.57 | 4.52   | 58.55 | 5.48   | 54.56 | 3.27 | 56.87 | <b>2.24</b> | <b>62.12</b> |
|                     | 50           | 81.92  | <b>61.32</b> | 11.60     | 51.89 | 4.56   | 56.97 | 4.34   | 53.15 | 3.33 | 58.02 | <b>2.37</b> | 59.97        |

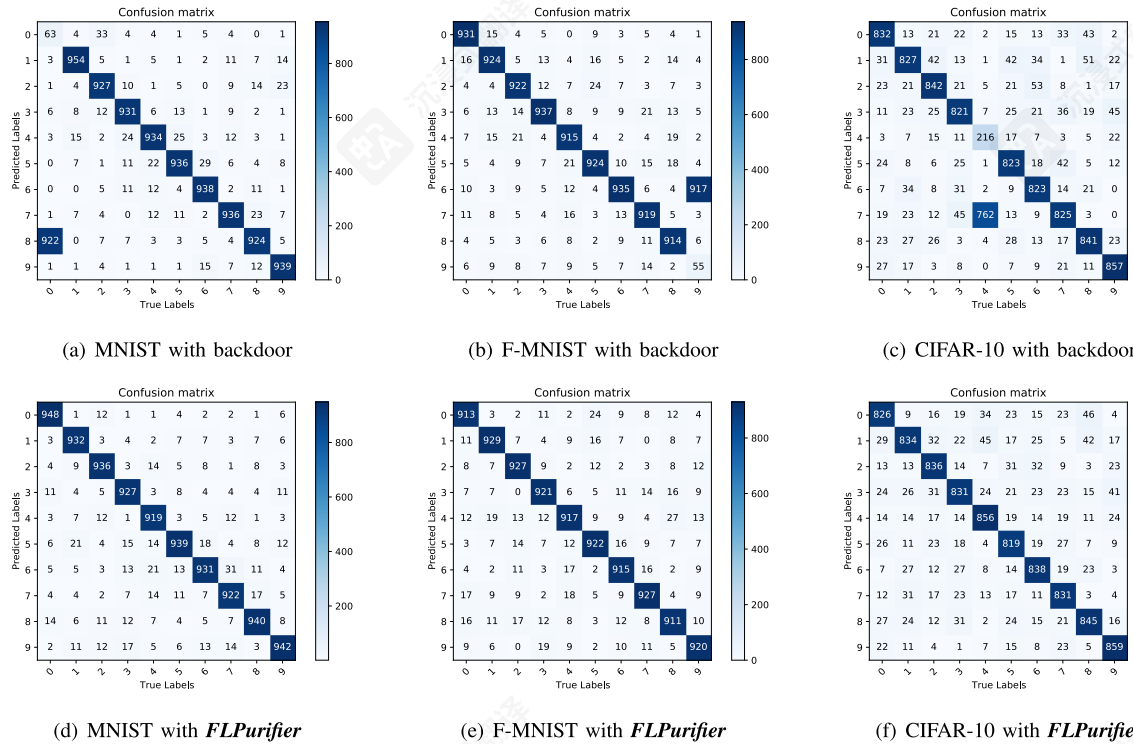


Fig. 4. Confusion matrix of the proposed *FLPurifier* and baseline backdoor model on MNIST, F-MNIST, and CIFAR-10 datasets.

weight classifier which represents the universal trend of classification, and then dynamically adjusts the weight according to the deviation between each classifier and it. Such settings reduce the weight of low-quality classifiers resulting from the limitation of the quality of the local data, and also reduce the weight of the backdoor classifier since the backdoor classifier also has a distance from the average classifier. In contrast, the increase in ASR is slight because the success of backdoor defense is mainly due that we break the connection between trigger features and the target label in the training process by decoupled contrastive training while the ACA just plays an auxiliary role. Moreover, comparing the IID and Non-IID settings, we can draw that ACA achieves better performance meaning that ACA is more needed in Non-IID settings since the data of each client in Non-IID settings are unbalanced and thus the local classifier trained on these data are divergent.

ACA can relieve these problems and improve the consistency of the classifiers.

3) *Different Ratios of Malicious Client*: To ensure the effectiveness and reliability of FL in the presence of backdoor attacks, it is crucial for defense methods to be robust to varying proportions of malicious participants. In this part, we aimed to assess the robustness of different defense methods against backdoor attacks at varying malicious participation ratios. Specifically, we set the malicious participation ratio  $r$  which represents the ratio of the number of malicious clients appearing in each round of FL to the total number of clients selected in that round. We investigated the MA and ASR of *FLPurifier* compared with four defense methods under three types of attacks. Note that the curve of Fedavg represents the result without any defense. Fig. 6 illustrates six subfigures of the trend of MA (the upper) and ASR (the lower)

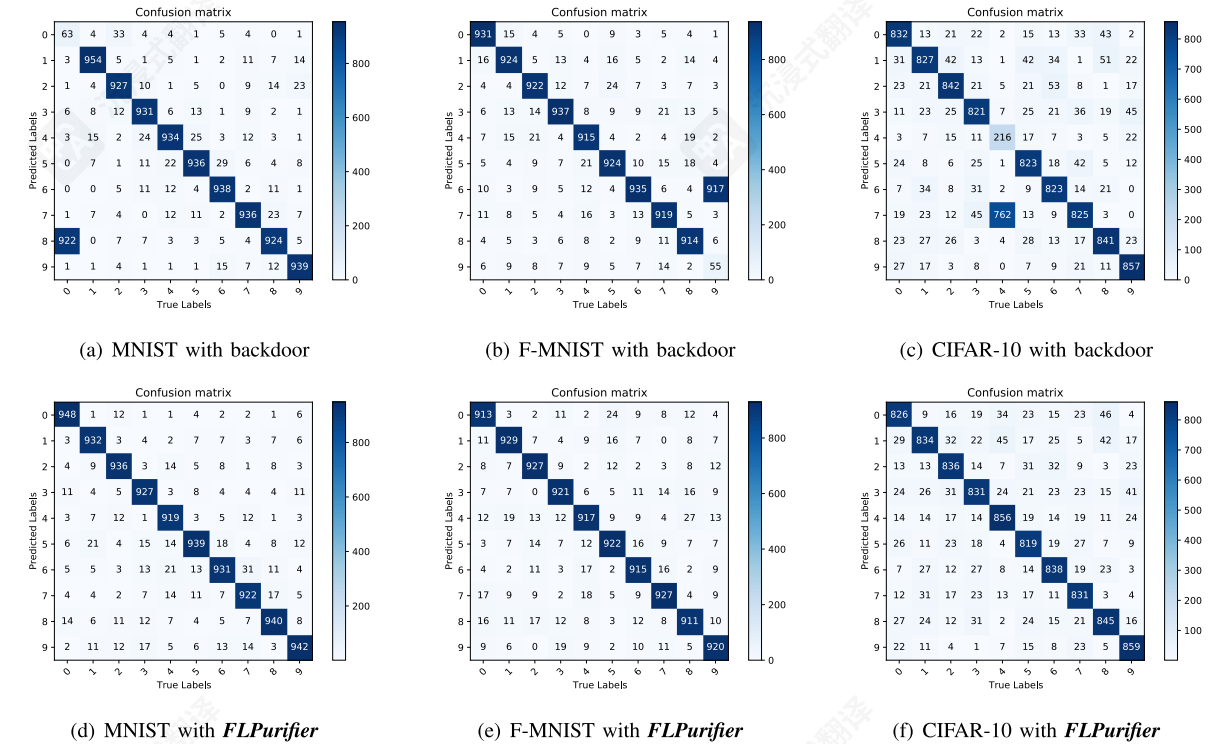
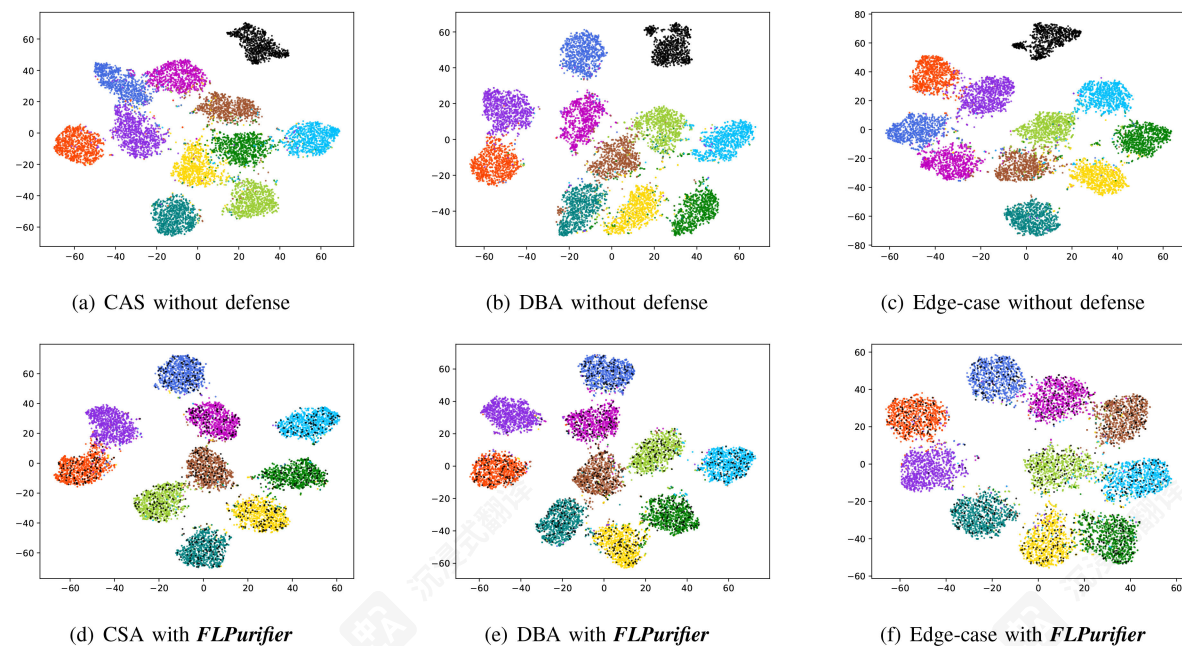
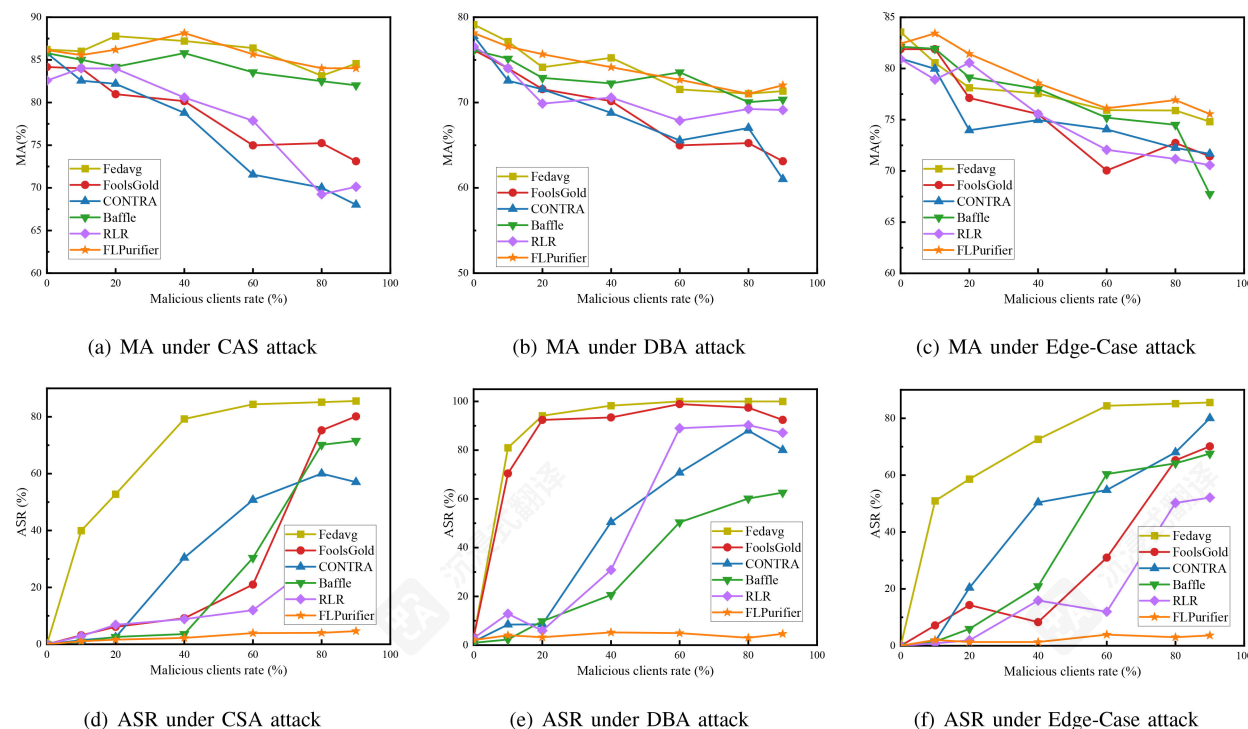


图 4. 在 MNIST、F-MNIST 和 CIFAR-10 数据集上，所提出的 *FLPurifier* 和基线后门模型之间的混淆矩阵。

权重分类器，它表示分类的普遍趋势，然后根据每个分类器与其之间的偏差动态调整权重。这种设置减少了由于本地数据质量限制而导致的低质量分类器的权重，同时也减少了后门分类器的权重，因为后门分类器与平均分类器之间也存在距离。相比之下，ASR 的增加很小，因为后门防御的成功主要归因于我们在训练过程中通过解耦对比训练打破了触发特征和目标标签之间的连接，而 ACA 仅起辅助作用。此外，比较 IID 和非 IID 设置，我们可以得出 ACA 实现了更好的性能，这意味着 ACA 在非 IID 设置中更受需要，因为非 IID 设置中每个客户端的数据是不平衡的，因此在这些数据上训练的本地分类器是发散的。

ACA 可以缓解这些问题并提高分类器的的一致性。

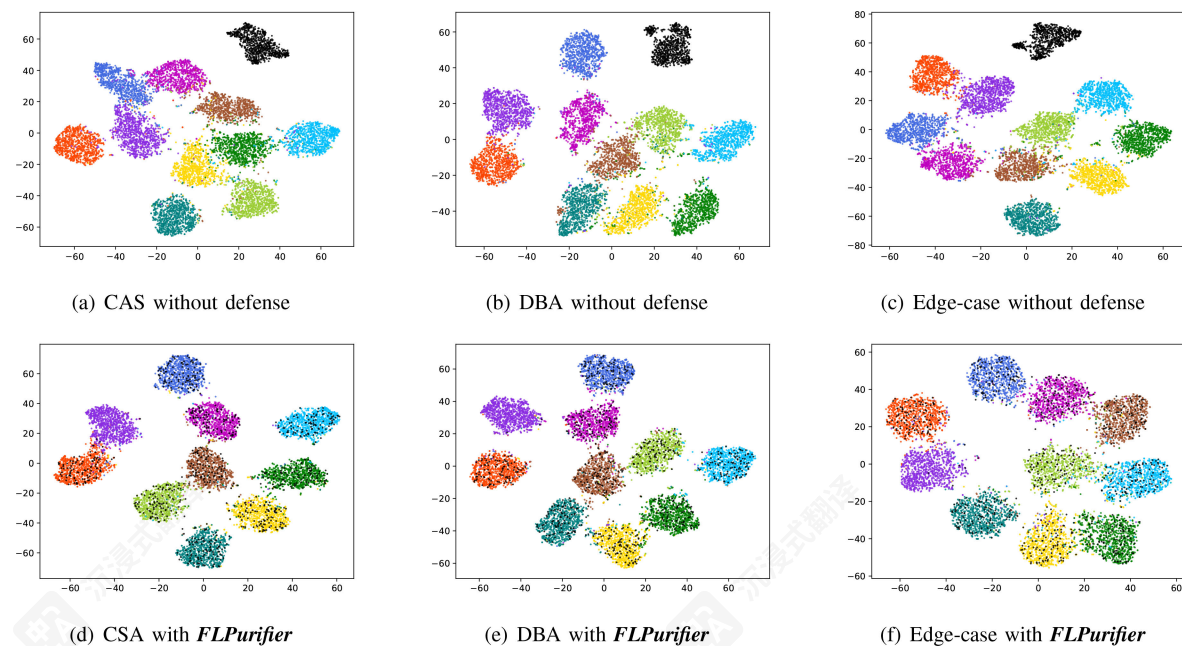
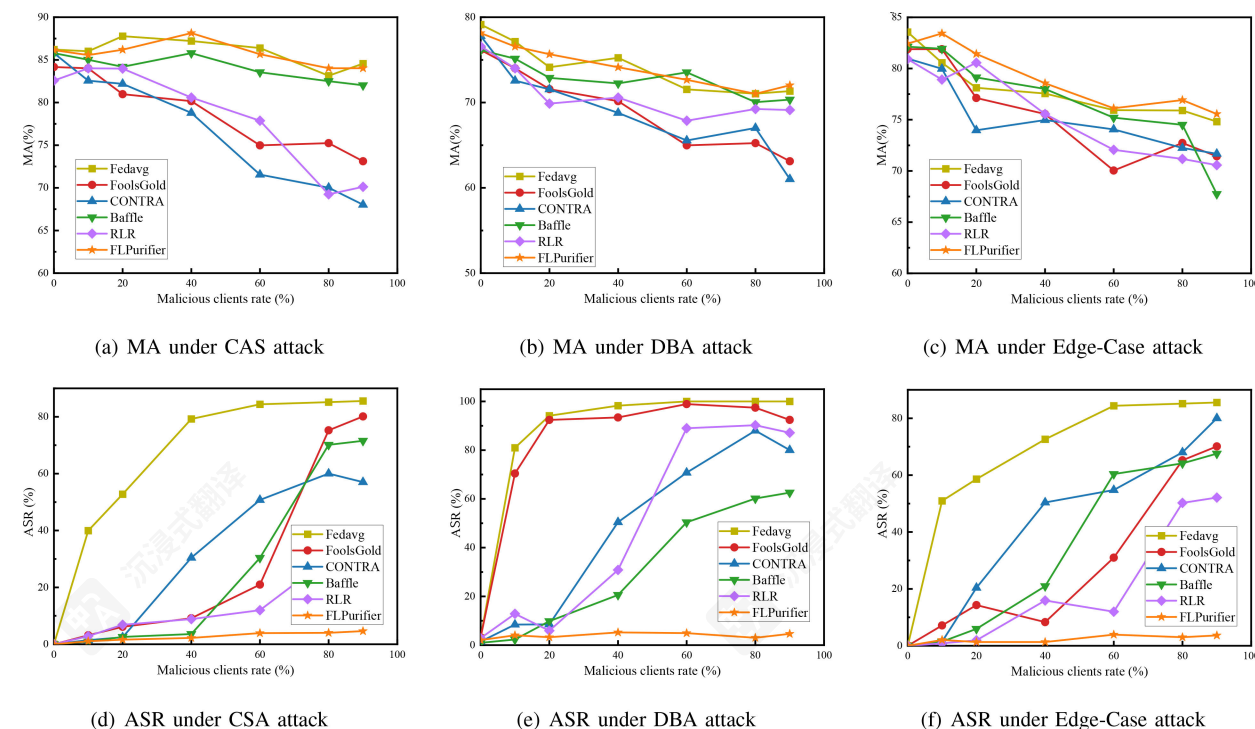
3) 恶意客户端的不同比例：为了确保在存在后门攻击的情况下联邦学习 (FL) 的有效性和可靠性，防御方法必须对恶意参与者的不同比例具有鲁棒性。在本部分中，我们旨在评估不同防御方法在恶意参与比例不同的情况下对后门攻击的鲁棒性。具体来说，我们设置了恶意参与比例  $r$ ，它表示在 FL 每一轮中出现的恶意客户端数量与该轮中选定的客户端总数的比例。我们研究了 *FLPurifier* 与四种防御方法在三种类型攻击下的 MA 和 ASR。请注意，Fedavg 的曲线表示没有任何防御的结果。图 6 展示了六个子图，分别展示了 MA（上部）和 ASR（下部）的趋势

Fig. 5. The t-SNE of the feature representations under different attacks on CIFAR-10 *with/without* *FLPurifier*.Fig. 6. The defense performance of *FLPurifier* against CAS, DBA, and Edge-Case attacks with the different rates of malicious clients on CIFAR-10 dataset.

with the increasing malicious participation ratio for a specific attack.

As we can observe that the MA shows a decreasing trend as the malicious participation ratio increase while the ASR appears a significant increase. It is reasonable because more malicious participants mean more backdoors are embedded, causing greater disturbance to the global model. Meanwhile, the increasing of malicious clients implies plenty of backdoor data are used to train the target model, which causes improvement in ASR. However, *FLPurifier* maintained relatively high

main task accuracy and effectively mitigated the success rate of backdoor attacks across different malicious participation ratios. On the contrary, baseline methods face various challenges at higher malicious participation ratios. FoolsGold defended the backdoor by reducing the weight of malicious participants. However, when the number of attackers is sufficient, even if their weight is reduced, it can still promote the success of backdoor attacks. Similarly, CONTRA implements a measure through a reputation scheme to determine the credibility of local model parameters to dynamically adjust the

Fig. 5. The t-SNE of the feature representations under different attacks on CIFAR-10 *with/without* *FLPurifier*.Fig. 6. *FLPurifier* 对 CAS、DBA 和 Edge-Case 攻击的防御性能, 以及在不同恶意客户端比例下 CIFAR-10 的表现 dataset.

随着特定攻击的恶意参与率增加。

如我们所见, 随着恶意参与率的增加, MA 呈下降趋势, 而 ASR 则出现显著增加。这是合理的, 因为更多的恶意参与者意味着嵌入的后门更多, 对全局模型造成更大的干扰。同时, 恶意客户端的增加意味着使用了更多的后门数据进行目标模型的训练, 这导致 ASR 的提升。然而, *FLPurifier* 保持了相对较高的主要任务精度, 并有效地降低了不同恶意参与率下的后门攻击成功率。相反, 基线方法在高恶意参与率下面临各种挑战。FoolsGold 通过降低恶意参与者的权重来防御后门。然而, 当攻击者数量足够多时, 即使他们的权重被降低, 仍然可以促进后门攻击的成功。类似地, CONTRA 通过声誉机制来决定本地模型参数的可信度, 以动态调整

恶意客户端的比例。FoolsGold 通过降低恶意参与者的权重来防御后门。然而, 当攻击者数量足够多时, 即使他们的权重被降低, 仍然可以促进后门攻击的成功。类似地, CONTRA 通过声誉机制来决定本地模型参数的可信度, 以动态调整



TABLE VI  
COMPARISON RESULTS OF DEFENSE PERFORMANCE WITH OR WITHOUT  
ACA ON IID SETTING

| Dataset   | Attack    | Without ACA |       | With ACA |       |
|-----------|-----------|-------------|-------|----------|-------|
|           |           | ASR         | MA    | ASR      | MA    |
| MNIST     | DBA       | 4.73        | 85.14 | 2.69     | 90.32 |
|           | Edge-Case | 1.59        | 89.21 | 1.47     | 92.57 |
|           | PGD       | 1.08        | 87.32 | 0.94     | 92.05 |
| F-MNIST   | DBA       | 4.76        | 86.16 | 2.31     | 90.19 |
|           | Edge-Case | 2.91        | 87.72 | 0.85     | 91.44 |
|           | PGD       | 1.96        | 88.14 | 1.26     | 91.89 |
| CIFAR-10  | CS        | 3.13        | 84.91 | 1.63     | 88.14 |
|           | DBA       | 4.75        | 70.59 | 3.24     | 75.66 |
|           | Edge-Case | 2.12        | 80.56 | 1.32     | 82.43 |
|           | PGD       | 3.02        | 76.54 | 1.58     | 79.21 |
| GTSRB     | CS        | 1.98        | 87.23 | 2.17     | 89.11 |
|           | DBA       | 3.16        | 74.65 | 2.50     | 89.92 |
|           | Edge-Case | 1.82        | 83.74 | 1.79     | 90.87 |
|           | PGD       | 3.61        | 74.98 | 1.56     | 89.27 |
| CIFAR-100 | CS        | 3.03        | 60.17 | 2.07     | 62.02 |
|           | DBA       | 3.74        | 56.13 | 2.81     | 62.36 |
|           | Edge-Case | 1.17        | 57.46 | 0.96     | 61.65 |
|           | PGD       | 2.09        | 54.54 | 1.41     | 60.18 |

TABLE VII  
COMPARISON RESULTS OF DEFENSE PERFORMANCE WITH OR WITHOUT  
ACA ON NON-IID SETTING

| Dataset | Setting         | Without ACA |       | With ACA |       |
|---------|-----------------|-------------|-------|----------|-------|
|         |                 | ASR         | MA    | ASR      | MA    |
| MNIST   | $\alpha = 0.05$ | 4.78        | 78.12 | 1.59     | 80.97 |
|         | $\alpha = 0.1$  | 6.04        | 81.77 | 3.04     | 86.72 |
|         | $\alpha = 10$   | 4.73        | 85.14 | 2.69     | 90.32 |
| E-MNIST | $\alpha = 0.05$ | 6.94        | 59.95 | 3.21     | 66.37 |
|         | $\alpha = 0.1$  | 5.98        | 63.32 | 3.56     | 69.68 |
|         | $\alpha = 10$   | 3.57        | 71.09 | 2.81     | 74.96 |

weight of each client, and the reputation scheme is ineffective under excessive malicious clients. Battle verifies whether there is a backdoor in the model through the private data of the client, and decides whether to abandon it through a vote strategy. Correspondingly, RLR determined the local update sign through voting. Obviously, when there are enough malicious participants, these mechanisms relying on voting are actually fatal. Instead, *FLPurifier* eliminates backdoor attacks from the root by decoupling the training process of encoders and classifiers, thereby being hardly affected in ASR by the ratios of malicious clients. However, the massive malicious clients cause limitations to ACA, which drops the MA slightly. Overall, the experiment results highlight the robustness of federated learning to different ratios of malicious clients.

4) *Different Attack Interval*: In certain scenarios, attackers may be randomly selected in each round, or the frequency of attacks may be reduced to enhance their concealment. Thus, we investigate the performance of *FLPurifier* under reduced or even zero attack frequencies. Specifically, we conduct experiments with different attack intervals to evaluate *FLPurifier* against DBA on the CIFAR-10 dataset. The experimental results, presented in Table VIII, demonstrate that

TABLE VIII  
PERFORMANCE OF *FLPurifier* UNDER DIFFERENT ATTACK INTERVAL

| Attack Interval | Before |       | <i>FLPurifier</i>         |                             |
|-----------------|--------|-------|---------------------------|-----------------------------|
|                 | MA     | ASR   | MA                        | ASR                         |
| Non-attack      | -      | -     | 84.57                     | -                           |
| 0               | 72.14  | 94.12 | 75.66 ( $\uparrow 3.23$ ) | 3.24 ( $\downarrow 90.88$ ) |
| 1               | 72.04  | 93.75 | 75.98 ( $\uparrow 3.27$ ) | 3.27 ( $\downarrow 90.48$ ) |
| 2               | 77.59  | 87.93 | 78.42 ( $\uparrow 3.01$ ) | 3.01 ( $\downarrow 84.92$ ) |
| 4               | 80.25  | 82.48 | 82.75 ( $\uparrow 2.56$ ) | 2.56 ( $\downarrow 79.92$ ) |
| 8               | 81.47  | 76.62 | 83.14 ( $\uparrow 1.79$ ) | 1.79 ( $\downarrow 74.83$ ) |
| 16              | 83.01  | 69.86 | 84.07 ( $\uparrow 1.21$ ) | 1.21 ( $\downarrow 68.65$ ) |

DBA maintains a relatively high attack success rate, despite a moderate decrease in performance with reduced attack frequencies. Furthermore, as the attack interval increases, the damage inflicted by DBA on MA is mitigated. As expected, *FLPurifier* also shows improved defense capabilities with lower attack frequencies. It is worth noting that even when deployed without any attacks, *FLPurifier* can maintain a high level of MA. These findings indicate that the attack frequency has no significant impact on the performance of *FLPurifier*, and the deployment of *FLPurifier* does not inflict harm on the federated learning process.

#### D. Applicability Analysis (RQ3)

1) *Number of Clients*: The number of clients is a key setting in the FL scenarios. To verify the applicability of our proposed method, we evaluate the impact on the performance of *FLPurifier* by varying the number of clients. Moreover, we extend the scaling factor  $S$  from 10 to 50 in the CAS attack and do experiments on six datasets to better evaluate the robustness of *FLPurifier*. Note that since MNIST, F-MNIST, and EMNIST are hard to find some nature feature to execute CAS backdoor attack, we set the scaling factor following the setting in CAS but evaluate against the DBA attack. In general, *FLPurifier* is effective in different client number settings. Although we can see from Fig. 7 that the success rate of backdoor attacks decreases with the increase in the number of clients, we think it is mainly because more clients decrease the aggregation weights of attackers. Additionally, *FLPurifier* is hardly affected by attack intensity since the amplified weight of the backdoor classifier is effectively alleviated by the adaptive aggregation strategy. This is because as the scaling factor increases, the difference between the backdoor classifier and the average classifier also amplifies, which is easily detected by ACA, leading to a decrease in the weight of the backdoor classifier.

2) *Non-IID Data Distributions*: The different Non-IID data distributions are another significant and realistic setting in the FL scenarios where data is distributed in Non-IID fashion among agents. Following prior art [53], we model Non-IID data distributions using a Dirichlet distribution  $\text{Dir}(\alpha)$ , where a smaller  $\alpha$  indicates higher data heterogeneity. Specifically, we vary  $\alpha$  from 0.05 to 0.1 and 10 on MNIST and EMNIST against DBA, where  $\alpha = 10$  represents IID. Table IX reports the final results for the Non-IID setting. The results indicate that *FLPurifier* provides an effective defense with negligible

表 VI 在 ONIID 设置下或无 ACA 时防御性能的比较结果

| Dataset   | Attack    | Without ACA |       | With ACA |       |
|-----------|-----------|-------------|-------|----------|-------|
|           |           | ASR         | MA    | ASR      | MA    |
| MNIST     | DBA       | 4.73        | 85.14 | 2.69     | 90.32 |
|           | Edge-Case | 1.59        | 89.21 | 1.47     | 92.57 |
|           | PGD       | 1.08        | 87.32 | 0.94     | 92.05 |
| F-MNIST   | DBA       | 4.76        | 86.16 | 2.31     | 90.19 |
|           | Edge-Case | 2.91        | 87.72 | 0.85     | 91.44 |
|           | PGD       | 1.96        | 88.14 | 1.26     | 91.89 |
| CIFAR-10  | CS        | 3.13        | 84.91 | 1.63     | 88.14 |
|           | DBA       | 4.75        | 70.59 | 3.24     | 75.66 |
|           | Edge-Case | 2.12        | 80.56 | 1.32     | 82.43 |
|           | PGD       | 3.02        | 76.54 | 1.58     | 79.21 |
| GTSRB     | CS        | 1.98        | 87.23 | 2.17     | 89.11 |
|           | DBA       | 3.16        | 74.65 | 2.50     | 89.92 |
|           | Edge-Case | 1.82        | 83.74 | 1.79     | 90.87 |
|           | PGD       | 3.61        | 74.98 | 1.56     | 89.27 |
| CIFAR-100 | CS        | 3.03        | 60.17 | 2.07     | 62.02 |
|           | DBA       | 3.74        | 56.13 | 2.81     | 62.36 |
|           | Edge-Case | 1.17        | 57.46 | 0.96     | 61.65 |
|           | PGD       | 2.09        | 54.54 | 1.41     | 60.18 |

表 VII 在 ONNON-IID 设置下或无 ACA 时防御性能的比较结果

| Dataset | Setting         | Without ACA |       | With ACA |       |
|---------|-----------------|-------------|-------|----------|-------|
|         |                 | ASR         | MA    | ASR      | MA    |
| MNIST   | $\alpha = 0.05$ | 4.78        | 78.12 | 1.59     | 80.97 |
|         | $\alpha = 0.1$  | 6.04        | 81.77 | 3.04     | 86.72 |
|         | $\alpha = 10$   | 4.73        | 85.14 | 2.69     | 90.32 |
| E-MNIST | $\alpha = 0.05$ | 6.94        | 59.95 | 3.21     | 66.37 |
|         | $\alpha = 0.1$  | 5.98        | 63.32 | 3.56     | 69.68 |
|         | $\alpha = 10$   | 3.57        | 71.09 | 2.81     | 74.96 |

每个客户端的权重, 并且当恶意客户端过多时, 声誉机制是无效的。Battle 通过客户端的私有数据验证模型中是否存在后门, 并通过投票策略决定是否放弃。相应地, RLR 通过投票确定本地更新的符号。显然, 当有足够的恶意参与者时, 这些依赖投票的机制实际上是致命的。相反, *FLPurifier* 通过解耦编码器和分类器的训练过程从根本上消除后门攻击, 因此在 ASR 中很少受恶意客户端比例的影响。然而, 大量的恶意客户端限制了 ACA, 导致 MA 略有下降。总体而言, 实验结果表明联邦学习对不同比例的恶意客户端具有鲁棒性。

4) 不同的攻击间隔: 在某些场景下, 攻击者可能在每一轮中被随机选择, 或者攻击频率可能被降低以增强其隐蔽性。因此, 我们研究了 *FLPurifier* 在降低甚至零攻击频率下的性能。具体来说, 我们通过不同的攻击间隔进行实验, 以评估 *FLPurifier* 在 CIFAR-10 数据集上对 DBA 的防御能力。实验结果, 如表 VIII 所示, 表明

表 VIII 不同的攻击间隔下 *FLPurifier* 的性能

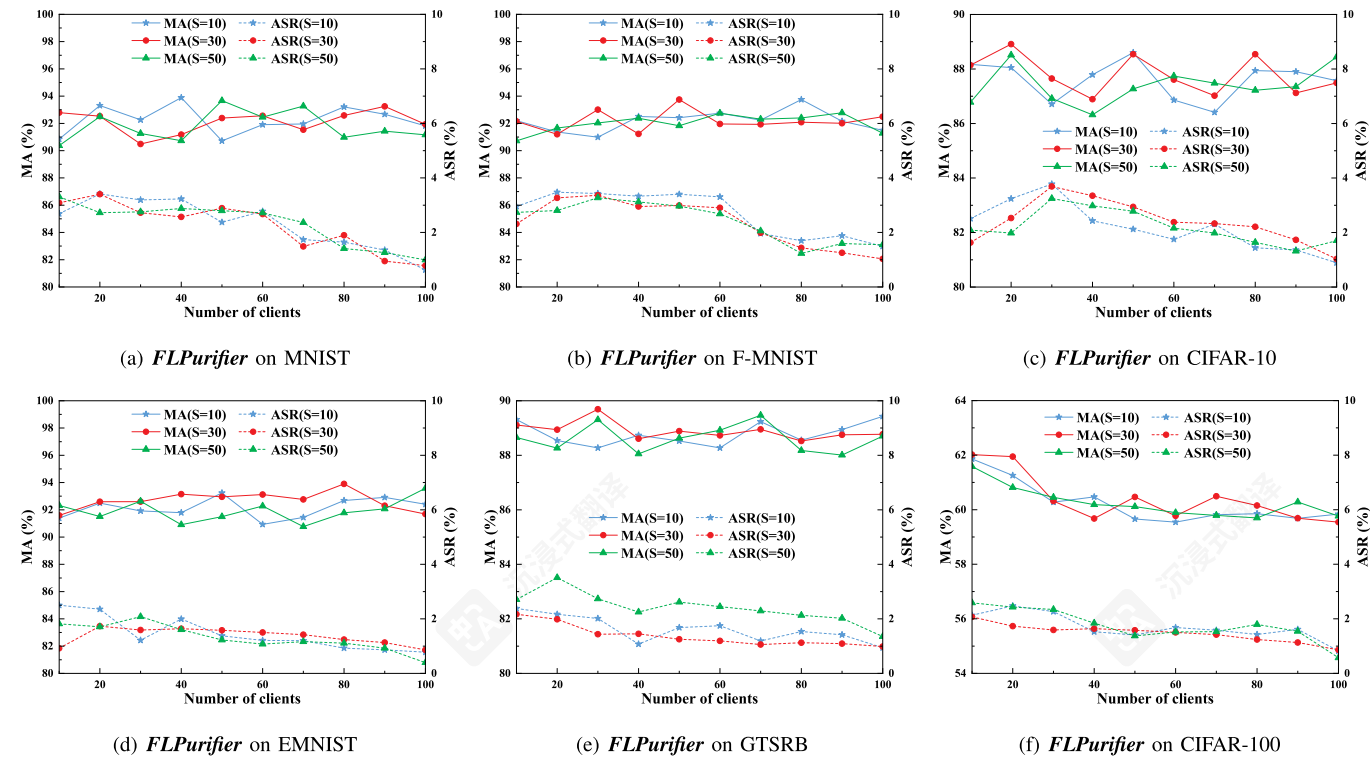
| Attack Interval | Before |       | <i>FLPurifier</i>         |                             |
|-----------------|--------|-------|---------------------------|-----------------------------|
|                 | MA     | ASR   | MA                        | ASR                         |
| Non-attack      | -      | -     | 84.57                     | -                           |
| 0               | 72.14  | 94.12 | 75.66 ( $\uparrow 3.23$ ) | 3.24 ( $\downarrow 90.88$ ) |
| 1               | 72.04  | 93.75 | 75.98 ( $\uparrow 3.27$ ) | 3.27 ( $\downarrow 90.48$ ) |
| 2               | 77.59  | 87.93 | 78.42 ( $\uparrow 3.01$ ) | 3.01 ( $\downarrow 84.92$ ) |
| 4               | 80.25  | 82.48 | 82.75 ( $\uparrow 2.56$ ) | 2.56 ( $\downarrow 79.92$ ) |
| 8               | 81.47  | 76.62 | 83.14 ( $\uparrow 1.79$ ) | 1.79 ( $\downarrow 74.83$ ) |
| 16              | 83.01  | 69.86 | 84.07 ( $\uparrow 1.21$ ) | 1.21 ( $\downarrow 68.65$ ) |

DBA 在攻击频率降低的情况下仍然保持相对较高的攻击成功率, 尽管其性能有所下降。此外, 随着攻击间隔的增加, DBA 对 MA 造成的损害得到缓解。正如预期的那样, *FLPurifier* 在攻击频率较低时也表现出更好的防御能力。值得注意的是, 即使在没有攻击的情况下, *FLPurifier* 也能保持较高的 MA 水平。这些结果表明, 攻击频率对 *FLPurifier* 的性能没有显著影响, 并且 *FLPurifier* 的部署不会对联邦学习过程造成损害。

#### D. 适用性分析 (RQ3)

1) 客户端数量: 客户端数量是联邦学习场景中的一个关键设置。为了验证我们提出的方法的适用性, 我们评估了改变客户端数量对 *FLPurifier* 性能的影响。此外, 我们在 CAS 攻击中将扩展因子  $S$  从 10 扩展到 50, 并在六个数据集上进行实验, 以更好地评估 *FLPurifier* 的鲁棒性。请注意, 由于 MNIST、F-MNIST 和 EMNIST 难以找到一些自然特征来执行 CAS 后门攻击, 我们按照 CAS 中的设置设置扩展因子, 但评估针对 DBA 攻击。总的来说, *FLPurifier* 在不同的客户端数量设置中都很有效。尽管我们可以从图 7 中看到, 随着客户端数量的增加, 后门攻击的成功率会降低, 但我们认为这主要是因为更多的客户端降低了攻击者的聚合权重。此外, *FLPurifier* 几乎不受攻击强度的影响, 因为后门分类器的放大权重通过自适应聚合策略得到了有效缓解。这是因为随着扩展因子的增加, 后门分类器与平均分类器之间的差异也会放大, 这很容易被 ACA 检测到, 从而导致后门分类器的权重降低。

2) 非独立同分布数据 (*Non-IID Data*) 分布: 不同的非独立同分布 (Non-IID) 数据分布是联邦学习场景中另一个重要且现实的设置, 其中数据在代理之间以非独立同分布 (Non-IID) 的方式分布。遵循现有技术 [53], 我们使用 Dirichlet 分布  $\text{Dir}(\alpha)$  对非独立同分布 (Non-IID) 数据分布进行建模, 其中较小的  $\alpha$  表示更高的数据异质性。具体来说, 我们在 MNIST 和 EMNIST 上对 DBA 变化  $\alpha$  从 0.05 到 0.1 和 10, 其中  $\alpha = 10$  表示独立同分布 (IID)。表 IX 报告了非独立同分布 (Non-IID) 设置的最终结果。结果表明 *FLPurifier* 提供了有效的防御, 且开销可忽略不计

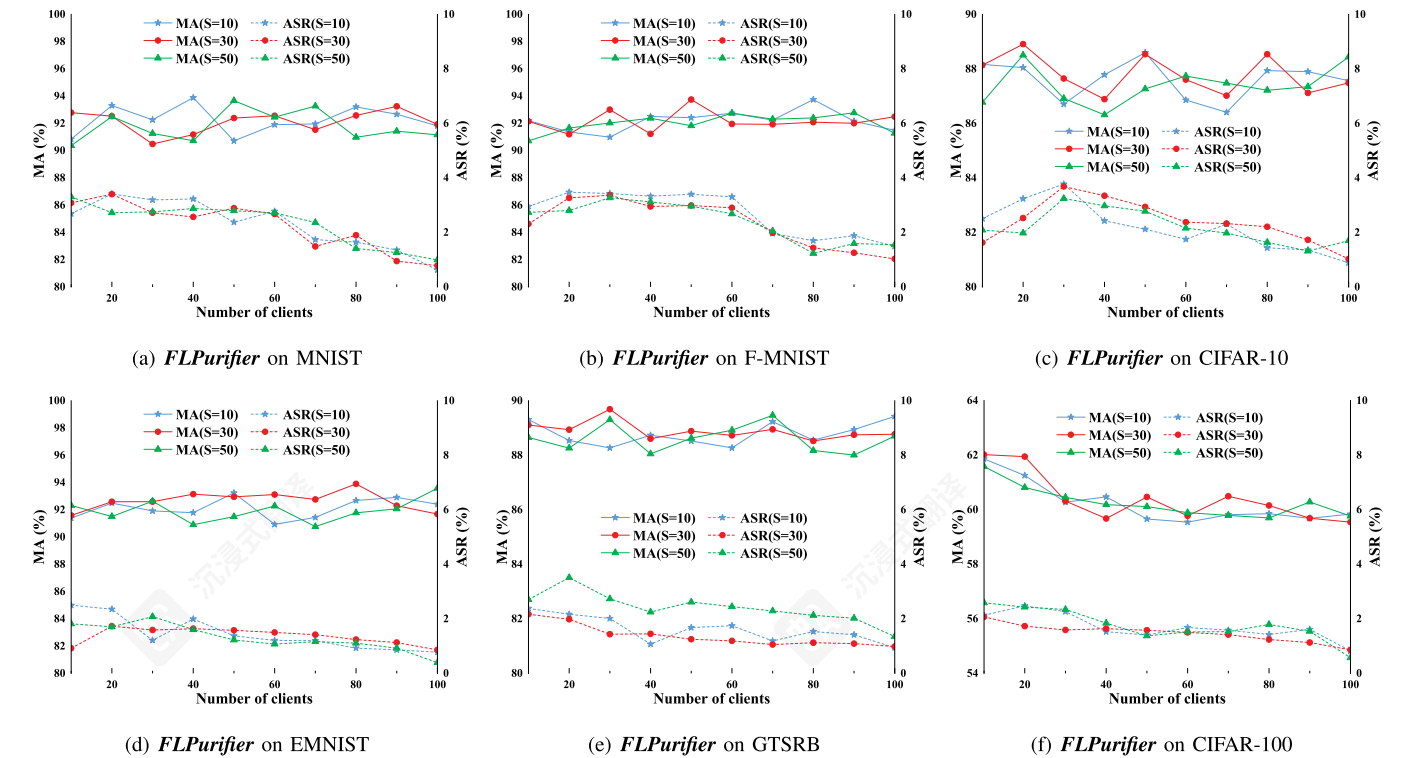
Fig. 7. The defense performance of *FLPurifier* against CAS attack with the different numbers of clients on six datasets.

degradation in the main task accuracy which outperforms other methods. In more detail, *FLPurifier* lowers the ASR of DBA to below 4 % under the Non-IID setting, while the baseline methods exhibit evident degradation, where only the RLR still retains unsatisfactory results (round 9%). Surprisingly, FoolsGold which is proved particularly effective in Non-IID has almost lost its effect in defending against DBA. For CONTRA, it measures the alignment level among clients and a higher alignment level means that the client's local objective is closer to the one of another client, which is more suspicious to be malicious. However, the alignment level is hard to measure since the data distribution is extremely imbalanced. Similarly, the imbalanced data distribution also enhances the difficulty of Battle in detecting malicious updates. In contrast, *FLPurifier* demonstrated excellent results by mitigating backdoors during the training process. Therefore, it can be drawn that *FLPurifier* is almost independent of data distribution and has excellent applicability.

3) *Adaptive Attacks*: As previously discussed, the success of backdoor attacks stems from the injection process where the attacker adds triggers to a normal sample and modifies its label. These trigger features form a distinct cluster associated with the target label under supervised training. To address this issue, our proposed approach involves decoupling the encoder and classifier structures of the model through contrastive training, thereby breaking the connection between backdoor features and target labels. Assume that the attacker is aware of our defense intuition and method. A feasible approach for them is to implant backdoor attacks without relying on attack-specific labels. In other words, they would employ untargeted or clean-label backdoor attacks. In this scenario, the attacker

directly adds triggers to a clean sample while maintaining its original label (e.g., “dog”). During the training phase, the attacker expects the model to learn the trigger features present in the sample and associate that the trigger belongs to the dog class. Consequently, in the prediction stage, the sample containing embedded triggers would be misclassified as “dog.” The fundamental nature of these attacks lies in establishing a connection between the trigger features and the normal features of the target class (the trigger features and the normal features of the target class form a cohesive cluster in the feature space), rather than establishing a direct connection between the trigger features and the target label (where the trigger features act as an individual cluster center). This idea aligns with previous invisible and clean-label backdoor attacks, thus we adopt this method as our adaptive attacker strategy.

As shown in Table X, *FLPurifier* still exhibits high performance by significantly reducing ASR to a mere 3% across all datasets. Furthermore, it maintains a negligible decrease in MA and even demonstrates slight improvements on complex datasets. These outcomes can be primarily attributed to the trick employed during the local training process, which is detailed in the methods section. In brief, *FLPurifier* leverages the local encoder of the last federated communication round as the target encoder while employing the global encoder as the online encoder. Despite adaptive attacks embedding a backdoor into the online encoder, resulting that the outputs of the online encoder for the samples with triggers are similar to that for the same samples without triggers due to the formation of a cohesive cluster between trigger features and normal features of the samples in the feature space. However, the target encoder lacks this characteristic. Consequently, the

Fig. 7. The defense 性能的 *FLPurifier* 对 CAS 攻击的对比

bers of clients on six datasets.

在主要任务精度上的退化，优于其他方法。更详细地，*FLPurifier* 将 DBA 的 ASR 降至 Non-IID 设置下的 4% 以下，而基线方法表现出明显的退化，其中只有 RLR 仍然保留不令人满意的结果（约 9%）。令人惊讶的是，FoolsGold 被证明在 Non-IID 中特别有效，但在防御 DBA 方面几乎失去了效果。对于 CONTRA，它测量客户端之间的对齐水平，较高的对齐水平意味着客户端的本地目标是另一个客户端目标更接近，这更可疑是恶意的。然而，由于数据分布极不平衡，对齐水平很难测量。同样，不平衡的数据分布也增加了 Battle 检测恶意更新的难度。相比之下，*FLPurifier* 通过在训练过程中缓解后门问题，展示了优异的结果。因此，可以得出结论，*FLPurifier* 几乎独立于数据分布，具有极好的适用性。

3) *自适应攻击*: 如前所述，后门攻击的成功源于攻击者在正常样本中注入触发器并修改其标签的过程。这些触发器特征在监督训练下与目标标签形成了一个独特的簇。为了解决这个问题，我们提出的方法通过对比训练将模型的编码器和分类器结构解耦，从而切断后门特征与目标标签之间的联系。假设攻击者了解我们的防御思路和方法。对他们来说，一个可行的方法是不依赖攻击特定标签的后门攻击。换句话说，他们会采用非目标或干净标签的后门攻击。在这种情况下，攻击者

直接在干净样本中添加触发器，同时保持其原始标签（例如，“狗”）。在训练阶段，攻击者期望模型学习样本中存在的触发器特征，并将其关联为触发器属于狗类。因此，在预测阶段，包含嵌入触发器的样本会被误分类为“狗”。这些攻击的根本性质在于建立触发器特征与目标类正常特征之间的联系（触发器特征和目标类正常特征在特征空间中形成一个紧密的簇），而不是建立触发器特征与目标标签之间的直接联系（触发器特征作为独立的簇中心）。这一思路与之前的不可见和干净标签后门攻击一致，因此我们采用这种方法作为我们的自适应攻击者策略。

如表 X 所示，*FLPurifier* 通过显著降低 ASR 至所有数据集的 3% 以下，仍然表现出高性能。此外，它保持了 MA 的微小下降，甚至在复杂数据集上表现出轻微的改进。这些结果主要归因于本地训练过程中采用的技巧，该技巧在方法部分进行了详细说明。简而言之，*FLPurifier* 利用最后一次联邦通信轮次的本地编码器作为目标编码器，同时将全局编码器作为在线编码器。尽管自适应攻击将后门嵌入在线编码器中，但由于触发特征和样本正常特征在特征空间中形成了紧密的簇，导致在线编码器对带有触发器的样本的输出与相同样本的无触发器输出相似。然而，目标编码器缺乏这一特性。因此，



TABLE IX

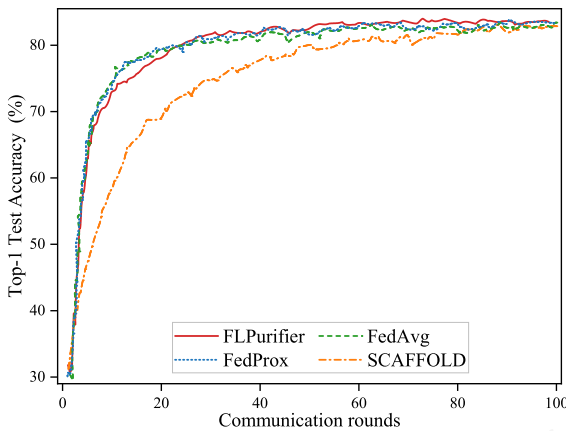
COMPARISON RESULTS OF DIFFERENT FEDERATED BACKDOOR DEFENSE METHODS ON MNIST AND EMNIST (NON-IID)

| Datasets | Setting         | Before |              | FoolsGold |       | CONTRA |       | Baffle |       | RLR   |       | FLPurifier  |              |
|----------|-----------------|--------|--------------|-----------|-------|--------|-------|--------|-------|-------|-------|-------------|--------------|
|          |                 | ASR    | MA           | ASR       | MA    | ASR    | MA    | ASR    | MA    | ASR   | MA    | ASR         | MA           |
| MNIST    | $\alpha = 0.05$ | 84.52  | <b>84.09</b> | 83.26     | 76.92 | 39.74  | 80.89 | 42.18  | 79.21 | 11.56 | 80.14 | <b>1.59</b> | 83.95        |
|          | $\alpha = 0.1$  | 90.63  | <b>87.85</b> | 89.43     | 78.81 | 24.56  | 82.62 | 28.16  | 80.17 | 9.15  | 81.59 | <b>3.04</b> | 86.83        |
|          | $\alpha = 10$   | 100    | <b>90.43</b> | 97.73     | 79.23 | 7.13   | 87.45 | 8.64   | 82.97 | 6.51  | 86.19 | <b>2.69</b> | 90.32        |
| EMNIST   | $\alpha = 0.05$ | 80.71  | <b>67.96</b> | 78.74     | 63.49 | 36.72  | 61.49 | 39.12  | 62.73 | 9.11  | 62.12 | <b>3.21</b> | 67.54        |
|          | $\alpha = 0.1$  | 85.67  | 72.42        | 80.55     | 67.66 | 35.31  | 64.53 | 40.12  | 64.16 | 9.24  | 63.39 | <b>3.56</b> | <b>73.05</b> |
|          | $\alpha = 10$   | 94.29  | <b>76.78</b> | 87.12     | 70.09 | 17.62  | 71.02 | 11.53  | 71.42 | 7.87  | 72.10 | <b>2.81</b> | 75.18        |

TABLE X

PERFORMANCE OF *FLPurifier* UNDER ADAPTIVE BACKDOOR ATTACK

| Attack   | Dataset   | Before |      | <i>FLPurifier</i> |      |
|----------|-----------|--------|------|-------------------|------|
|          |           | ASR    | MA   | ASR               | MA   |
| Adaptive | MNIST     | 96.3   | 92.8 | 2.6               | 91.6 |
|          | F-MNIST   | 95.8   | 91.9 | 3.1               | 89.5 |
|          | EMNIST    | 94.2   | 75.7 | 1.7               | 75.4 |
|          | CIFAR-10  | 97.1   | 83.7 | 0.9               | 84.1 |
|          | GTSRB     | 96.9   | 89.5 | 2.5               | 88.3 |
|          | CIFAR-100 | 91.8   | 59.7 | 1.4               | 60.8 |

Fig. 8. The defense performance of *FLPurifier* compared with different aggregation methods on CIFAR-10 dataset.

target encoder exerts a corrective effect on the online encoder. Upon completing the training, the backdoor features are effectively eliminated. These findings affirm the effectiveness of our method in countering adaptive attacks.

#### E. Overhead of *FLPurifier* (RQ4)

Figure 8 illustrates the accuracy achieved in each round during training by comparing *FLPurifier* with three SOTA FL frameworks, i.e., Fedavg [1], FedProx [54], and SCAFFOLD [55]. Note that no attackers were included in these experiments. According to Figure 8, we can observe that the speed of accuracy improvement in *FLPurifier* is slightly lower compared with FedAvg and FedProx in the initial rounds (before 20 rounds). However, they achieve convergence and even higher accuracy in nearly the same number of rounds. This observation suggests that *FLPurifier* does not incur additional communication overhead compared to traditional

TABLE XI

COMMUNICATION AND COMPUTATION COST OF *FLPurifier*

| Target Accuracy | Communication (rounds) |     |      |      | Computation (epochs) |     |      |      |
|-----------------|------------------------|-----|------|------|----------------------|-----|------|------|
|                 | E=2                    | E=6 | E=10 | E=20 | E=2                  | E=6 | E=10 | E=20 |
| 70%             | 60                     | 30  | 15   | 8    | 120                  | 180 | 150  | 160  |
| 80%             | 70                     | 45  | 30   | 20   | 140                  | 270 | 300  | 400  |

federated learning algorithms. However, during the server aggregation phase, there is a computational cost associated with calculating the similarity between different local models. This overhead, although present, is deemed acceptable considering the need for maintaining security in federated learning settings, particularly when mobile phones, which often serve as clients in FL, require robust security guarantees but possess limited computational resources. On the other hand, the server typically possesses substantial computational capabilities, making it better suited to handle such overhead efficiently.

Table XI presents a comprehensive analysis of the necessary communication rounds and computation epochs for achieving a target accuracy using different local epochs  $E$  which is the sum of  $E_{con}$  and  $E_{sup}$ . Increasing  $E$  reduces the communication cost as it decreases the required rounds, but it generally leads to higher computation costs. For example, consider the scenario where  $E = 6$ , which necessitates 45 rounds of communication to achieve 80% accuracy with 270 computation epochs. On the other hand, if we set  $E = 20$ , only 8 rounds of communication are required, but it demands 400 computation epochs. These findings highlight the inherent trade-off between the communication cost and the computation cost in the context of federated learning.

#### V. SUMMARY AND FUTURE WORK

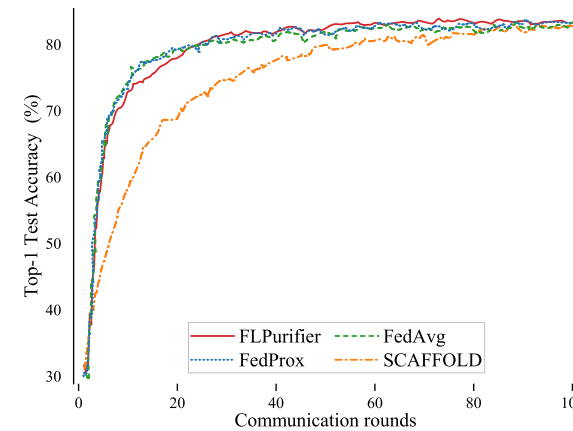
In this paper, we proposed *FLPurifier*, a novel backdoor defense framework in FL based on supervised contrastive training. Specifically, *FLPurifier* instructs clients to separately train a clean feature extractor by contrastive learning and a classifier in a supervising manner, thus breaking the connection between backdoor features and target labels. To further eliminate the impact of malicious classifiers and improve the accuracy of the global model, we present an adaptive classifier aggregation strategy, dynamically adjusting the weight coefficients of each classifier according to their deviation degree. Compared with existing defense methods, *FLPurifier*

表 IX 不同联邦后门防御方法在 MNIST 和 EMNIST (非独立同分布) 上的比较结果

| Datasets | Setting         | Before |              | FoolsGold |       | CONTRA |       | Baffle |       | RLR   |       | FLPurifier  |              |
|----------|-----------------|--------|--------------|-----------|-------|--------|-------|--------|-------|-------|-------|-------------|--------------|
|          |                 | ASR    | MA           | ASR       | MA    | ASR    | MA    | ASR    | MA    | ASR   | MA    | ASR         | MA           |
| MNIST    | $\alpha = 0.05$ | 84.52  | <b>84.09</b> | 83.26     | 76.92 | 39.74  | 80.89 | 42.18  | 79.21 | 11.56 | 80.14 | <b>1.59</b> | 83.95        |
|          | $\alpha = 0.1$  | 90.63  | <b>87.85</b> | 89.43     | 78.81 | 24.56  | 82.62 | 28.16  | 80.17 | 9.15  | 81.59 | <b>3.04</b> | 86.83        |
|          | $\alpha = 10$   | 100    | <b>90.43</b> | 97.73     | 79.23 | 7.13   | 87.45 | 8.64   | 82.97 | 6.51  | 86.19 | <b>2.69</b> | 90.32        |
| EMNIST   | $\alpha = 0.05$ | 80.71  | <b>67.96</b> | 78.74     | 63.49 | 36.72  | 61.49 | 39.12  | 62.73 | 9.11  | 62.12 | <b>3.21</b> | 67.54        |
|          | $\alpha = 0.1$  | 85.67  | 72.42        | 80.55     | 67.66 | 35.31  | 64.53 | 40.12  | 64.16 | 9.24  | 63.39 | <b>3.56</b> | <b>73.05</b> |
|          | $\alpha = 10$   | 94.29  | <b>76.78</b> | 87.12     | 70.09 | 17.62  | 71.02 | 11.53  | 71.42 | 7.87  | 72.10 | <b>2.81</b> | 75.18        |

表 X 在自适应后门攻击下 *FLPurifier* 的性能

| Attack   | Dataset   | Before |      | <i>FLPurifier</i> |      |
|----------|-----------|--------|------|-------------------|------|
|          |           | ASR    | MA   | ASR               | MA   |
| Adaptive | MNIST     | 96.3   | 92.8 | 2.6               | 91.6 |
|          | F-MNIST   | 95.8   | 91.9 | 3.1               | 89.5 |
|          | EMNIST    | 94.2   | 75.7 | 1.7               | 75.4 |
|          | CIFAR-10  | 97.1   | 83.7 | 0.9               | 84.1 |
|          | GTSRB     | 96.9   | 89.5 | 2.5               | 88.3 |
|          | CIFAR-100 | 91.8   | 59.7 | 1.4               | 60.8 |

图 8. *FLPurifier* 与不同聚合方法在 CIFAR-10 数据集上的防御性能比较。

目标编码器对在线编码器产生纠正效果。完成训练后, 后门特征被有效消除。这些发现证实了我们的方法在对抗自适应攻击方面的有效性。

#### E. *FLPurifier* 的开销 (RQ4)

图 8 展示了在训练过程中每一轮所达到的准确率, 通过比较 *FLPurifier* 与三个 SOTA FL 框架, 即 Fedavg [1], FedProx [54], 和 SCAFFOLD [55]。请注意, 这些实验中未包含攻击者。根据图 8, 我们可以观察到, 在初始轮次 (20 轮之前) 中, *FLPurifier* 的准确率提升速度略低于 FedAvg 和 FedProx。然而, 它们在几乎相同的轮次数内实现了收敛, 甚至达到了更高的准确率。这一观察结果表明, *FLPurifier* 与传统方法相比, 不会产生额外的通信开销

表 XI C 通信和 C 计算 C 成本 *FLPurifier*

| Target Accuracy | Communication (rounds) |     |      |      | Computation (epochs) |     |      |      |
|-----------------|------------------------|-----|------|------|----------------------|-----|------|------|
|                 | E=2                    | E=6 | E=10 | E=20 | E=2                  | E=6 | E=10 | E=20 |
| 70%             | 60                     | 30  | 15   | 8    | 120                  | 180 | 150  | 160  |
| 80%             | 70                     | 45  | 30   | 20   | 140                  | 270 | 300  | 400  |

联邦学习算法。然而, 在服务器聚合阶段, 计算不同本地模型之间的相似性存在计算开销。这种开销虽然存在, 但在需要维护联邦学习环境中的安全性的情况下被认为是可接受的, 特别是在移动设备 (通常作为 FL 中的客户端) 需要强大的安全保证但计算资源有限的情况下。另一方面, 服务器通常拥有强大的计算能力, 使其能够更有效地处理这种开销。

Table XI presents a comprehensive analysis of the necessary communication rounds and computation epochs for achieving a target accuracy using different local epochs  $E$  which is the sum of  $E_{con}$  and  $E_{sup}$ . Increasing  $E$  reduces the communication cost as it decreases the required rounds, but it generally leads to higher computation costs. For example, consider the scenario where  $E = 6$ , which necessitates 45 rounds of communication to achieve 80% accuracy with 270 computation epochs. On the other hand, if we set  $E = 20$ , only 8 rounds of communication are required, but it demands 400 computation epochs. These findings highlight the inherent trade-off between the communication cost and the computation cost in the context of federated learning.

#### V. SUMMARY AND FUTURE WORK

In this paper, we proposed *FLPurifier*, a novel backdoor defense framework in FL based on supervised contrastive training. Specifically, *FLPurifier* instructs clients to separately train a clean feature extractor by contrastive learning and a classifier in a supervising manner, thus breaking the connection between backdoor features and target labels. To further eliminate the impact of malicious classifiers and improve the accuracy of the global model, we present an adaptive classifier aggregation strategy, dynamically adjusting the weight coefficients of each classifier according to their deviation degree. Compared with existing defense methods, *FLPurifier*

can eliminate the negative impacts of backdoor attack in the training procedure. The extensive experiments validate that *FLPurifier* is effective against backdoor attacks and outperforms the existing defense. In future work, we will investigate how to deploy *FLPurifier* in the scenario of heterogeneous or personalized federated learning.

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可以消除训练过程中的后门攻击的负面影响。广泛的实验验证 *FLPurifier* 对后门攻击是有效的, 并且优于现有的防御。在未来的工作中, 我们将研究如何将 *FLPurifier* 部署在异构或个性化联盟学习的场景中。

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federated learning, data aggregation, and collaborated learning. He served as an editor for journal of CSI and program committee for more than 30 international conferences.



**Chunpeng Ge** (Member, IEEE) received the Ph.D. degree in computer science from Nanjing University of Aeronautics and Astronautics in 2016. He was a Research Fellow with the University of Wollongong and Singapore University of Technology and Design. He has published more than 60 papers in prestigious journal and conferences. His current research interests include information security and privacy-preserving for cloud computing, blockchain, and security and privacy of AI systems. His recent work focused on the topics of cryptographic tools for aggregation, and collaborated learning. He served as an editor for journal of CSI and program committee for more than 30 international conferences.

**Bing Chen** (Member, IEEE) received the B.S. and M.S. degrees in computer engineering from Nanjing University of Aeronautics and Astronautics (NUAA), China, in 1992 and 1995, respectively, and the Ph.D. degree from the College of Compute Science and Technology, NUAA, in 2008. He is currently a Full Professor with the College of Compute Science and Technology, NUAA. His research interests include cloud/edge computing, security and privacy, FL, and wireless communications.



**Jiale Zhang** (Member, IEEE) received the Ph.D. degree in computer science and technology from the College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2021. He is currently an Associate Professor with the School of Information Engineering, Yangzhou University, Yangzhou, China. He has published over 40 research papers in refereed international conferences and journals, such as IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, IEEE INTERNET OF THINGS JOURNAL, COSE, JSS, IEEE ICC, and IEEE Globecom. His research interests are mainly include federated learning, AI security, and blockchian security.



include software maintenance and evolution, software repository mining, and intelligence analysis.

**Chengcheng Zhu** received the B.S. degree from Yangzhou University, Yangzhou, China, in 2022, where he is currently pursuing the M.S. degree in computer science and technology. His research interests include backdoor attacks and adversarial learning.

**Xiaobing Sun** (Member, IEEE) received the B.S. degree in computer science and technology from Jiangsu University of Science and Technology, Zhenjiang, China, in 2007, and the Ph.D. degree from the School of Computer Science and Engineering, Southeast University, Nanjing, China, in 2012. He is currently a Professor with the School of Information Engineering, Yangzhou University, Yangzhou, China. He has been authorized more than 20 patents. He has published more than 80 articles in refereed international journals. His research interests



**Willy Susilo** (Fellow, IEEE) received the Ph.D. degree in computer science from the University of Wollongong, Australia. He is currently a Distinguished Professor and the Head of the School of Computing and Information Technology and the Director of the Institute of Cybersecurity and Cryptology, University of Wollongong. He has published over 400 research papers in the area of cybersecurity and cryptology. His main research interests include cybersecurity, cryptography, and information security. He was a recipient of the Australian Research Council (ARC) Future Fellow by the ARC and the Researcher of the Year Award by the University of Wollongong in 2016. He has served as a program committee member for dozens of international conferences. He is the Editor-in-Chief of the *Information* journal. He is currently serving as an Associate Editor for several international journals, including the *Computer Standards and Interface* (Elsevier) and the *International Journal of Information Security* (Springer). His work has been cited over 16000 times in Google Scholar.



**Shui Yu** (Fellow, IEEE) received the Ph.D. degree from Deakin University, Australia, in 2004. He is a Professor of the School of Computer Science; and the Deputy Chair of the University Research Committee, University of Technology Sydney, Australia. He has published five monographs and edited two books, more than 500 technical papers at different venues, such as IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING, IEEE TRANSACTIONS ON PARALLEL AND DISTRIBUTED SYSTEMS, IEEE TRANSACTIONS ON COMPUTERS, IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, IEEE TRANSACTIONS ON MOBILE COMPUTING, IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING, IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING, IEEE/ACM TRANSACTIONS ON NETWORKING, and INFOCOM. His current H-index is 74. He has promoted the research field of networking for big data since 2013, and his research outputs have been widely adopted by industrial systems, such as Amazon Cloud Security. His research interests include cybersecurity, network science, big data, and mathematical modeling. He is a member of ACM and AAAS. He is currently serving on the editorial boards of IEEE COMMUNICATIONS SURVEYS AND TUTORIALS (an Area Editor) and IEEE INTERNET OF THINGS JOURNAL (an Editor). He served as a Distinguished Lecturer for IEEE Communications Society, from 2018 to 2021. He is a Distinguished Visitor of IEEE Computer Society and an Elected Member of Board of Governors of IEEE VTS and ComSoc.

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**Jiale Zhang** (IEEE 会员) 于 2021 年获得南京航空航天大学计算机科学与技术学院博士学位。他目前是扬州大学信息工程学院副教授。他在 IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS、IEEE INTERNET OF THINGS JOURNAL 等国际会议和期刊上发表了 40 余篇研究论文, 如 IEEE ICC 和 IEEE Globecom。他的研究兴趣主要包括联邦学习、AI 安全和区块链安全。



包括软件维护与演进、软件仓库挖掘和智能分析。

**Chengcheng Zhu** 于2022 年获得扬州大学学士学位, 目前在该校攻读计算机科学与技术硕士学位。他的研究兴趣包括后门攻击和对抗学习。

**孙晓冰** (IEEE 会员) 于 2007 年毕业于江苏科技大学, 获计算机科学与技术学士学位, 2012 年毕业于东南大学计算机科学与工程学院, 获博士学位。他目前是扬州大学信息工程学院教授。他已获得 20 多项专利授权。他在国际期刊上发表了 80 多篇论文。他的研究兴趣



联邦学习、数据聚合和协作学习。他担任过 CSI 期刊的编委和 30 多个国际会议的程序委员会成员。



**葛春鹏** (IEEE 会员) 于 2016 年毕业于南京航空航天大学, 获计算机科学博士学位。他曾担任伍伦贡大学和新加坡南洋理工大学设计学院的研究员。他在知名期刊和会议上发表了 60 多篇论文。他当前的研究兴趣包括云计算、区块链和人工智能系统的信息安全和隐私保护。他的近期工作集中在密码学工具方面

**Bing Chen** (IEEE 会员) 于 1992 年和 1995 年分别获得南京航空航天大学 (NUAA) 计算机工程专业的学士和硕士学位, 并于 2008 年获得 NUAA 计算机科学与技术学院的博士学位。他目前是 NUAA 计算机科学与技术学院的正教授。他的研究兴趣包括云 / 边缘计算、安全与隐私、FL 和无线通信。



**Willy Susilo** (IEEE Fellow) 于澳大利亚伍伦贡大学获得计算机科学博士学位。他目前是伍伦贡大学 computing and Information Technology 学院的杰出教授和院长, 以及网络安全与密码学研究所的所长。他在网络安全与密码学领域发表了 400 多篇研究论文。他的主要研究兴趣包括网络安全、密码学和信息安全。他曾获得澳大利亚研究奖学金



**水宇** (IEEE Fellow) 于 2004 年从澳大利亚迪肯大学获得博士学位。他是计算机科学学院的教授; 以及澳大利亚悉尼科技大学大学研究委员会的副主席。他出版了五本专著, 编辑了两本书, 在不同的场合发表了 500 多篇技术论文, 例如 IEEE《可靠与安全计算汇刊》、《并行与分布式系统汇刊》、《计算机汇刊》等。

IEEE 信息取证与安全汇刊, IEEE 移动计算汇刊, IEEE 知识与数据工程汇刊, IEEE 新兴计算领域汇刊, IEEE/ACM 网络技术汇刊和 INFOCOM. 他目前的 H 指数是 74。自 2013 年以来, 他一直推动大数据网络领域的研究, 他的研究成果已被工业系统广泛采用, 例如亚马逊云安全。他的研究兴趣包括网络安全、网络科学、大数据和数学建模。他是 ACM 和 AAAS 的会员。他目前担任 IEEE 通信综述与教程 (担任区域编辑) 和 IEEE 物联网汇刊 (担任编辑) 的编委。他于 2018 年至 2021 年担任 IEEE 通信学会杰出讲员。他是 IEEE 计算机学会的杰出访问学者, 以及 IEEE VTS 和 ComSoc 理事会当选成员。